

Principal series for stack of Coherent sheaves on curve

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1 Geometric and categorical players

1.1 Stacks of coherent sheaves and G -torsors on curve

We let X be a connected smooth projective curve over $\mathbf{k} = \mathbb{C}$ or $\overline{\mathbb{F}}_q$.

Recall that for $\mathcal{Y}$ a stack over $\mathbf{k}$, the mapping stack $\mathbf{Maps}(X, \mathcal{Y})$ is a prestack in groupoids: $\mathbf{Maps}(X, \mathcal{Y}) : \text{Sch}_{\mathbf{k}} \rightarrow \text{Gpd}; S \mapsto \mathcal{Y}(X \times S)$. For any affine group scheme H , we denote by $\mathbf{Bun}_H$ the mapping stack $\mathbf{Maps}(X, \text{pt}/H) = \mathbf{Maps}(X, \mathbf{B}H)$. It is well-known that $\mathbf{Bun}_H$ is an algebraic stack.

The coherent sheaves and vector bundles on X can be classified by Euler class, namely a pair of rank and degree. For $\alpha \in \mathbf{Z}^+ := \{(r, d) \mid r > 1 \text{ or } r = 0, d \geq 0\}$, we let $\mathbf{Coh}_{\alpha}$ be the stack of coherent sheaves of class α on X and $\mathbf{Bun}_{\alpha}$ the stack of vector bundles on X when $\alpha \in \mathbf{Z}^{++} := \mathbb{Z}_{\geq 1} \times \mathbb{Z}$, it is a connected component of $\mathbf{Bun}_{\text{GL}_{\text{rk}(\alpha)}} = \bigsqcup_{d \in \mathbb{Z}} \mathbf{Bun}_{(\text{rk}(\alpha), d)}$ and it is also open in $\mathbf{Coh}_{\alpha}$. When $\alpha \in \mathbf{Z}^+$ is of rank 0 and degree d , we use $\mathbf{Coh}_{\tau d}$ to denote $\mathbf{Coh}_{\alpha}$.

Proposition 1.1. *For any $\alpha \in \mathbf{Z}^{++}$, $\mathbf{Bun}_{\alpha} \subseteq \mathbf{Coh}_{\alpha}$ is a open substack, both of them are smooth, irreducible of dimension $(\text{rk}(\alpha))^2(g-1)$. We denote by $j_{\alpha} : \mathbf{Bun}_{\alpha} \hookrightarrow \mathbf{Coh}_{\alpha}$.*

More than that, we can determine a stratification over the complement of $\mathbf{Bun}_{\alpha}$ in $\mathbf{Coh}_{\alpha}$. We know any coherent sheaf $\mathcal{E}$ on X can be decomposed uniquely into torsion-free and torsion parts: $\mathcal{E}^{\text{tor.free}}$ and $\mathcal{E}^{\text{tor}}$, this gives us a stratification of $\mathbf{Coh}_{\alpha}$. For $\alpha \in \mathbf{Z}^+, d \geq 0$, we denote by $\mathbf{Coh}_{\alpha}^{\text{tor.deg}=d}$ (resp. $\mathbf{Coh}_{\alpha}^{\text{tor.deg} \leq d}$) the substack whose points are coherent sheaf of class α and degree of its torsion part is d (resp. $\leq d$). Then $\mathbf{Coh}_{\alpha} = \bigsqcup_{d \geq 0} \mathbf{Coh}_{\alpha}^{\text{tor.deg}=d}$ and $\mathbf{Coh}_{\alpha}^{\text{tor.deg} \leq d} = \bigsqcup_{0 \leq d' \leq d} \mathbf{Coh}_{\alpha}^{\text{tor.deg}=d'}$.

Proposition 1.2. (1) *For any $d \geq 0$, $\mathbf{Coh}_{\alpha}^{\text{tor.deg} \leq d}$ is open in $\mathbf{Coh}_{\alpha}$. We have a series of open exhaustion:*

$$\mathbf{Bun}_{\alpha} = \mathbf{Coh}_{\alpha}^{\text{tor.deg}=0} \hookrightarrow \mathbf{Coh}_{\alpha}^{\text{tor.deg} \leq 1} \hookrightarrow \dots \mathbf{Coh}_{\alpha} = \mathbf{Coh}_{\alpha}^{\text{tor.deg} \leq \infty}$$

(2) *For any d , $\mathbf{Coh}_{\alpha}^{\text{tor.deg}=d}$ is a gerbe of unipotent group of rank $-\text{rk}(\alpha) \cdot d$ over $\mathbf{Bun}_{\alpha-\tau d} \times \mathbf{Coh}_{\tau d}$.*

Let $\mathcal{E}$ be a coherent sheaf on X with class (r, d) , its slope is defined to be $\mu(\mathcal{E}) = \frac{d}{r}$, if $r = 0$, we put $\mu(\mathcal{E}) = \infty$. For any coherent sheaf $\mathcal{E} \in \mathbf{Coh}_{\alpha}(\mathbf{k})$, it admits a unique filtration $\mathcal{E}_1 \subseteq \mathcal{E}_2 \subseteq \dots \mathcal{E}_s = \mathcal{E}$ such that the slopes $\mu(\mathcal{E}_{i+1}/\mathcal{E}_i) \leq \mu(\mathcal{E}_i/\mathcal{E}_{i-1})$ called the *Harder-Narasimhan filtration*. In fact, this filtration exists for coherent sheaves on curve X_0 defined over $\overline{\mathbb{F}}_q$ and is stable under field extensions.

For any partition $\underline{\alpha} = (\alpha_1, \dots, \alpha_s)$ of α , we say it is of Harder-Narasimhan (HN in short) type if $\mu(\alpha_i) \geq \mu(\alpha_{i+1})$. We denote by $\text{HN}(\alpha)$ the set of all partitions of α that are HN type. There is a partial order on $\text{HN}(\alpha)$, for $\underline{\alpha}_1, \underline{\alpha}_2 \in \text{HN}(\alpha)$, we say $\underline{\alpha}_1 \leq \underline{\alpha}_2$ if the corresponding HN polygon of $\underline{\alpha}_1$ is beyond the one of $\underline{\alpha}_2$. There is a stratification on $\mathbf{Coh}_{\alpha}$ labelled by $(\text{HN}(\alpha), \leq)$, we denote each stratum by $\text{HN}^{-1}(\underline{\alpha})$, classifying those in $\mathbf{Coh}_{\alpha}$ with HN type $\underline{\alpha}$. Each $\text{HN}^{-1}(\underline{\alpha})$ is locally closed, with closure $\overline{\text{HN}^{-1}(\underline{\alpha})} = \bigsqcup_{\underline{\beta} \leq \underline{\alpha}} \text{HN}^{-1}(\underline{\beta})$. For later use, we divide the Harder-Narasimhan types into two kinds.

Notice for $\underline{\beta} = (\beta_1, \dots, \beta_s)$ with $\mu(\beta_i) - \mu(\beta_{i+1}) > 2g - 2$, the Harder-Narasimhan filtration splits and thus any $\mathcal{E} \in \text{HN}^{-1}(\underline{\beta})(\mathbf{k})$ can be written as direct sum $\mathcal{F}_1 \oplus \dots \oplus \mathcal{F}_s$ with each $\mathcal{F}_i$ of class β_i . We call such kind of HN types *very unstable*. For those HN types such that $\mu(\beta_i) - \mu(\beta_{i+1}) \leq 2g - 2$ for any i , we call them *almost semi-stable*. The rest of HN types are mixture of very unstable ones and almost semi-stable ones, we call them *not very unstable*. Those three types of locus in $\mathbf{Coh}_\alpha$ are denoted respectively by

$$\mathbf{Coh}_\alpha^{\text{vu}} \quad \mathbf{Coh}_\alpha^{\text{as}} \quad \mathbf{Coh}_\alpha^{\text{nvu}}$$

Another exhaustion of $\mathbf{Coh}_\alpha$ is given by global quotient stack of the form $[\mathcal{X}/\mathcal{G}]$ and is used by Sala-Schiffmann in [SS20] to construct K-theoretical Hall algebras. Fix a line bundle $\mathcal{L}$ over X , we say a coherent sheaf $\mathcal{F}$ is *strongly generated* by $\mathcal{L}$ if $\text{Hom}(\mathcal{L}, \mathcal{F}) \otimes \mathcal{L} \rightarrow \mathcal{F}$ is surjective and $\text{Ext}^1(\mathcal{L}, \mathcal{F}) = \{0\}$. For an Euler class α , we denote by $\mathbf{Coh}_\alpha^{>\mathcal{L}} \subseteq \mathbf{Coh}_\alpha$ the substack classifies the coherent sheaves that strongly generated by $\mathcal{L}$.

Proposition 1.3. [Sch16, Lemma 6.1; SS20, Proposition 2.1] (1) Let $\mathcal{L}$ be a line bundle on X with Euler class $c(\mathcal{L})$, then $\mathbf{Coh}_\alpha^{>\mathcal{L}}$ is an open substack in $\mathbf{Coh}_\alpha$. Moreover, $\mathbf{Coh}_\alpha^{>\mathcal{L}}$ is isomorphic to the quotient stack $[\mathcal{Q}_\alpha^\mathcal{L}/\mathcal{G}_\alpha^\mathcal{L}]$, where $\mathcal{Q}_\alpha^\mathcal{L}$ is a smooth and reduced subscheme in some Quot scheme and $\mathcal{G}_\alpha^\mathcal{L} \cong \text{GL}_{\langle c(\mathcal{L}), \alpha \rangle}$. (2) Consider the partial order on $\mathbf{Pic}_X$ so that another line bundle $\mathcal{L}' \geq \mathcal{L}$ if it is strongly generated by $\mathcal{L}$, then there is an open inclusion $j^{\mathcal{L}', \mathcal{L}} : \mathbf{Coh}_\alpha^{>\mathcal{L}'} \hookrightarrow \mathbf{Coh}_\alpha^{>\mathcal{L}}$ and $\mathbf{Coh}_\alpha$ can be exhibited as $\text{colim}_{\mathcal{L} \in \mathbf{Pic}_X} \mathbf{Coh}_\alpha^{>\mathcal{L}}$.

Proof. For fixed $\mathcal{L}, \alpha$, we can first consider the Quot scheme $\text{Quot}_\alpha^\mathcal{L} := \text{Quot}_X(\mathcal{L} \otimes \mathbf{k}^{\langle \mathcal{L}, \alpha \rangle}, \alpha)$ that classifies surjections $\phi : \mathcal{L} \otimes \mathbf{k}^{\langle c(\mathcal{L}), \alpha \rangle} \twoheadrightarrow \mathcal{F}$ where $\mathcal{F} \in \mathbf{Coh}_\alpha$. We can define an open subscheme $\mathcal{Q}_\alpha^\mathcal{L} \subseteq \text{Quot}_\alpha^\mathcal{L}$ so that

$$\mathcal{Q}_\alpha^\mathcal{L}(\mathbf{k}) = \left\{ \left[\phi : \mathcal{L} \otimes \mathbf{k}^{\langle c(\mathcal{L}), \alpha \rangle} \twoheadrightarrow \mathcal{F} \right] \in \text{Quot}_\alpha^\mathcal{L} \mid \phi_* : \mathbf{k}^{\langle c(\mathcal{L}), \alpha \rangle} \xrightarrow{\cong} \text{Hom}(\mathcal{L}, \mathcal{F}) \right\}.$$

There is a natural action of $\mathcal{G}_\alpha^\mathcal{L}(\mathbf{k}) := \text{GL}_{\langle c(\mathcal{L}), \alpha \rangle}$ on $\mathbf{k}^{\langle c(\mathcal{L}), \alpha \rangle}$ that gives arises to an action of $\mathcal{G}_\alpha^\mathcal{L}$ on $\mathcal{Q}_\alpha^\mathcal{L}$. In particular, it follows from definition of strongly generation that $\mathbf{Coh}_\alpha^{>\mathcal{L}} \cong [\mathcal{Q}_\alpha^\mathcal{L}/\mathcal{G}_\alpha^\mathcal{L}]$. Moreover, $\mathcal{Q}_\alpha^\mathcal{L}$ is smooth as for any $\phi \in \mathcal{Q}_\alpha^\mathcal{L}$, we have the vanishing $\text{Ext}^1(\text{Ker}(\phi), \mathcal{F}) = 0$. An explicit computation of the cotangent complex $\mathbb{L}_{\mathbf{Coh}_\alpha^{>\mathcal{L}}}$ could be found in [SS20].

For (ii), given $\mathcal{L}' > \mathcal{L}$ consider the open subscheme $\mathcal{Q}_\alpha^{\mathcal{L}, \mathcal{L}'} \subseteq \mathcal{Q}_\alpha^\mathcal{L}$ contains those $[\phi : \mathcal{L} \otimes \mathbf{k}^{\langle c(\mathcal{L}), \alpha \rangle} \twoheadrightarrow \mathcal{F}]$ with $\mathcal{F} \in \mathbf{Coh}^{>\mathcal{L}'}$. Then we have $\mathbf{Coh}_\alpha^{>\mathcal{L}'} \cong [\mathcal{Q}_\alpha^{\mathcal{L}, \mathcal{L}'}/\mathcal{G}_\alpha^\mathcal{L}]$. Moreover, in [SS20], an canonical isomorphism $[\mathcal{Q}_\alpha^{\mathcal{L}, \mathcal{L}'}/\mathcal{G}_\alpha^\mathcal{L}] \xrightarrow{\cong} [\mathcal{Q}_\alpha^{\mathcal{L}'}/\mathcal{G}_\alpha^{\mathcal{L}'}]$ is constructed, so we get an canonical open embedding $j^{\mathcal{L}, \mathcal{L}'}$. ■

1.2 Categories of sheaves and miraculous duality

As in [Ari+22], let Y be a quasi-compact scheme over $\mathbf{k}$, denote by $\text{Shv}_{\text{constr}}(Y)$ the category of

- Constructible sheaves on the topological space $Y(\mathbb{C})$ when $\mathbf{k} = \mathbb{C}$.

- Holonomic $\mathcal{D}$ -modules on Y when $\text{char}(\mathbf{k}) = 0$.
- Constructible ℓ -adic sheaves on Y , where $\ell \neq \text{char}(\mathbf{k}) > 0$.

There is a perverse t -structure on $\text{Shv}_{\text{constr}}(Y)$, whose heart $\text{Shv}_{\text{constr}}(Y)^\heartsuit$ is the category of perverse sheaves on Y , denoted by $\text{Perv}(Y)$. We have theorem of Beilinson saying that $D^b(\text{Perv}(Y)) \cong \text{Perv}(Y)$. We denote by $\text{Shv}(Y)$ the ind-completion $\text{Shv}(Y) := \text{Ind}(\text{Shv}_{\text{constr}}(Y))$, on which the perverse t -structure on $\text{Shv}_{\text{constr}}(Y)$ can be extended to $\text{Shv}(Y)$ in the way that compatible with filtered colimits. So we can see the heart $\text{Shv}(Y)^\heartsuit$ of the extended perverse t -structure on ind-completion is $\text{Ind}(\text{Perv}(Y))$, the ind-completion of the heart $\text{Perv}(Y)$.

Now for any prestack X , we can define $\text{Shv}_{\text{constr}}(X)$ and $\text{Shv}(X)$ as $\varinjlim_{S \rightarrow X} \text{Shv}_{\text{constr}}(S)$ and $\varinjlim_{S \rightarrow X} \text{Shv}(S)$ respectively. Let $\text{Perv}(X)$ be the heart of perverse t -structure in $\text{Shv}(X)$.

We will use $D(X)$ to denote either $\text{Shv}(X)$ as above or $\mathcal{D}\text{-mod}(X)$, the whole ∞ -category of $\mathcal{D}$ -modules (not necessarily holonomic) when the base field is $\mathbb{C}$. Then we have $D(\mathbf{Bun}_G) \cong \varinjlim_{U \in \text{op-qc}(\mathbf{Bun}_G)} D(U)$ and $D(\mathbf{Coh}_r) := \varinjlim_{U \in \text{op-qc}(\mathbf{Coh}_r)} D(U)$, where $\text{op-qc}(-)$ denotes the set of quasi-compact opens. For $j_{1,2} : U_1 \hookrightarrow U_2$ inclusion of quasi-compact opens in $\mathbf{Bun}_G$ or $\mathbf{Coh}_r$, the limit is taken as $(j_{1,2})^* : D(U_2) \hookrightarrow D(U_1)$. Note that when we take $D(X)$ to be $\text{Shv}(X)$, it is always compactly generated, since we have well-definedness of six functors for $\text{Shv}(X)$. But in general $\mathcal{D}\text{-mod}(X)$ is not compactly generated for arbitrary (pre)stack X . For the stacks we are mostly interested in, we have the following:

Theorem 1.4. *Both $D(\mathbf{Bun}_G)$ and $D(\mathbf{Coh}_\alpha)$ are compactly generated hence dualizable.*

Proof. The compact generation of $D(\mathbf{Bun}_G)$ is main theorem in [DG15]. We now prove for $D(\mathbf{Coh}_\alpha)$. It suffices to prove that the quasi-compact opens $\{\mathbf{Coh}_\alpha^{>\mathcal{L}}\}$ are co-truncative. In other words, it suffices to show that the left adjoint $j_!^\mathcal{L}$ to $(j^\mathcal{L})^*$ exists for open inclusion $j^\mathcal{L} : \mathbf{Coh}_\alpha^{>\mathcal{L}} \hookrightarrow \mathbf{Coh}_\alpha$. Notice that $j^\mathcal{L}$ factors as $\mathbf{Coh}_\alpha^{>\mathcal{L}} \cong [\mathbf{Q}_\alpha^\mathcal{L}/G_\alpha^\mathcal{L}] \xrightarrow{i^\mathcal{L}} [\text{Quot}_\alpha^\mathcal{L}/G_\alpha^\mathcal{L}] \xrightarrow{\widetilde{j}^\mathcal{L}} \mathbf{Coh}_\alpha$, where $\widetilde{j}^\mathcal{L}$ sends $(\phi : \mathcal{L} \otimes \mathbf{k}^{(c(\mathcal{L}), \alpha)} \twoheadrightarrow \mathcal{F}) \mapsto \mathcal{F}$ is $G_\alpha^\mathcal{L}$ -equivariant, thus $\text{Quot}_\alpha^\mathcal{L} \rightarrow \mathbf{Coh}_\alpha$ factors through $\widetilde{j}^\mathcal{L}$. From the properness of $\text{Quot}_\alpha^\mathcal{L}$ and the fact that $\mathbf{Coh}_\alpha$ is locally of finite type, we know $\widetilde{j}^\mathcal{L}$ is proper and thus $\widetilde{j}_!^\mathcal{L}$ is well-defined. It remains to show that $i^\mathcal{L} : [\mathbf{Q}_\alpha^\mathcal{L}/G_\alpha^\mathcal{L}] \hookrightarrow [\text{Quot}_\alpha^\mathcal{L}/G_\alpha^\mathcal{L}]$ is co-truncative. We can reduce to the similar situation as in [DG15, Proposition 9.2.2] ■

In general, assume $\mathbf{C} = \text{Ind}(\mathbf{C}^c)$ is compactly generated, then it is dualizable, with dual category identified with $\mathbf{C}^\vee = \text{Ind}((\mathbf{C}^c)^{\text{op}})$. We denote by $\text{ev}_\mathbf{C} : \mathbf{C} \otimes \mathbf{C}^\vee \rightarrow \mathbf{Vect}$ the counit map, which is given by $\text{ev}_\mathbf{C}(\mathbf{c} \otimes \xi) = \text{Hom}_{\mathbf{C}^\vee}(\mathbf{c}^\vee, \xi)$.

In our case, the dual category of $D(\mathbf{Bun}_G)$ resp. $D(\mathbf{Coh}_r)$ in DGCat can be described as follows:

We denote by $D(\mathbf{Bun}_G)_{\text{co}}$ and $D(\mathbf{Coh}_r)_{\text{co}}$ the $*$ -generated category of sheaves/ $\mathcal{D}$ -modules on $\mathbf{Bun}_G$ and $\mathbf{Coh}_r$ respectively. More precisely, they are the colimits taken in cocomplete DG categories and

continuous functors as follows

$$D(\mathbf{Bun}_G)_{\text{co}} := \text{colim}_{U \in \text{op-qc}(\mathbf{Bun}_G)} D(U); \quad D(\mathbf{Coh}_r)_{\text{co}} := \text{colim}_{U \in \text{op-qc}(\mathbf{Coh}_r)} D(U)$$

For $j_{1,2} : U_1 \hookrightarrow U_2$, we have the corresponding colimit is taken as $(j_{1,2})_* : D(U_1) \hookrightarrow D(U_2)$. Since for any quasi-compact U , we have self-duality $D(U) \cong (D(U))^\vee$ (by Verdier duality), we can see that

$$\begin{aligned} \text{Maps}_{\text{DGCat}}(D(\mathbf{Bun}_G)_{\text{co}}, \mathbf{Vect}) &= \text{Maps}_{\text{DGCat}}(\text{colim}_{U \in \text{op-qc}(\mathbf{Bun}_G)} D(U), \mathbf{Vect}) \\ &= \lim_{U \in \text{op-qc}} \text{Maps}_{\text{DGCat}}(D(U), \mathbf{Vect}) \cong \lim_{U \in \text{op-qc}} D(U)^\vee \cong D(\mathbf{Bun}_G) \end{aligned}$$

The same for $\mathbf{Coh}_r$, we have $\text{Maps}_{\text{DGCat}}(D(\mathbf{Coh}_r)_{\text{co}}, \mathbf{Vect}) \cong D(\mathbf{Coh}_r)$. As a corollary of Theorem 1.4, both $D(\mathbf{Bun}_G)$ and $D(\mathbf{Coh}_r)$ are dualizable and compactly generated. Hence we can deduce that the dual categories are

$$D(\mathbf{Bun}_G)_{\text{co}} \cong D(\mathbf{Bun}_G)^\vee, \quad D(\mathbf{Coh}_r)_{\text{co}} \cong D(\mathbf{Coh}_r)^\vee$$

Also, the above identification gives us canonical pairings:

$$\langle -, - \rangle_{\mathbf{Bun}_G} : D(\mathbf{Bun}_G) \times D(\mathbf{Bun}_G)_{\text{co}} \longrightarrow \mathbf{Vect}; \quad \langle -, - \rangle_{\mathbf{Coh}_r} : D(\mathbf{Coh}_r) \times D(\mathbf{Coh}_r)_{\text{co}} \longrightarrow \mathbf{Vect}$$

In general, for any stack $\mathcal{Y}$ such that $D(\mathcal{Y})$ is dualizable, we have $\text{Maps}_{\text{DGCat}}(D(\mathcal{Y})^\vee, D(\mathcal{Y})) \cong D(\mathcal{Y}) \otimes (D(\mathcal{Y})^\vee)^\vee \cong D(\mathcal{Y} \times \mathcal{Y})$, which says elements in $D(\mathcal{Y} \times \mathcal{Y})$ give rises to a functor $D(\mathcal{Y})^\vee \longrightarrow D(\mathcal{Y})$. In particular, we will denote by $\text{Ps-Id}_{\mathcal{Y},!}$ and $\text{Ps-Id}_{\mathcal{Y},*}$ the functors correspond to $\Delta_!(\mathcal{Y})$ and $\Delta_*(\mathcal{Y})$ respectively, where $\Delta : \mathcal{Y} \rightarrow \mathcal{Y} \times \mathcal{Y}$ is the diagonal. If $\mathcal{Y}$ is quasi-compact, then $\text{Ps-Id}_{\mathcal{Y},*}$ is an equivalence, which fails when $\mathcal{Y}$ is no longer quasi-compact.

Definition 1.5. We say $\mathcal{Y}$ is **miraculous** if $\text{Ps-Id}_{\mathcal{Y},!}$ is an equivalence.

Theorem 1.6. Both $\mathbf{Bun}_G$ and $\mathbf{Coh}_r$ are miraculous.

Proof. The proof for $\mathbf{Coh}_r$ goes exactly the same as in [Gai17]. We sketch here the idea we decompose $D(\mathbf{Coh}_\alpha)$ into two parts $D(\mathbf{Coh}_\alpha)_{\text{cusp}}$ and $D(\mathbf{Coh}_\alpha)_{\text{eis}}$, such that the former is left orthogonal to the latter. Similarly for the dual category $D(\mathbf{Coh}_\alpha)_{\text{co}}$, we have orthogonal decomposition $D(\mathbf{Coh}_\alpha)_{\text{co,cusp}}$ and $D(\mathbf{Coh}_\alpha)_{\text{co,eis}}$. We can reduce the Theorem to three results: First, $\text{Ps-Id}_!$ and Ps-Id_* agree on $D(\mathbf{Coh}_\alpha)_{\text{co,cusp}}$ up to some shift. Second, there is a compatibility between $\text{Ps-Id}_!$ and Eis^{Coh} . Last, $\text{Ps-Id}_!$ is fully-faithful. We postpone the proof to next section. $\blacksquare$

2 Eisenstein series and constant term functors

Let $\text{Eis}_{\alpha,\beta}^{\text{Coh}} = \left(\text{p}_{\alpha,\beta}^{\text{Coh}} \right)_! \left(\text{q}_{\alpha,\beta}^{\text{Coh}} \right)^* [\text{shift}]$ and $\text{CT}_{\alpha,\beta}^{\text{Coh}} = \left(\text{q}_{\alpha,\beta}^{\text{Coh}} \right)_* \left(\text{p}_{\alpha,\beta}^{\text{Coh}} \right)^! [\text{shift}]$ along

$$\mathbf{Coh}_{\alpha,\beta} = \mathbf{Coh}_{\alpha} \times \mathbf{Coh}_{\beta} \xleftarrow{\text{q}_{\alpha,\beta}^{\text{Coh}}} \widetilde{\mathbf{Coh}}_{\alpha,\beta} \xrightarrow{\text{p}_{\alpha,\beta}^{\text{Coh}}} \mathbf{Coh}_{\alpha+\beta}$$

where $\widetilde{\mathbf{Coh}}_{\alpha,\beta}$ is the stack classifies subsheaves $\mathcal{E}_1 \subseteq \mathcal{E}_2$ that $\mathcal{E}_1$ is of class α and $\mathcal{E}_2$ is of class $\alpha + \beta$, the $[\text{shift}]$ comes from the relative dimension of $\text{q}_{\alpha,\beta}$ computed in Proposition 2.1. More generally, for a sequence of classes $\underline{\alpha} = (\alpha_1, \dots, \alpha_k)$, we can define $\text{Eis}_{\underline{\alpha}}^{\text{Coh}}$ and $\text{CT}_{\underline{\alpha}}^{\text{Coh}}$ to be pull-push along

$$\mathbf{Coh}_{\underline{\alpha}} = \mathbf{Coh}_{\alpha_1} \times \dots \times \mathbf{Coh}_{\alpha_k} \xleftarrow{\text{q}_{\underline{\alpha}}^{\text{Coh}}} \widetilde{\mathbf{Coh}}_{\underline{\alpha}} \xrightarrow{\text{p}_{\underline{\alpha}}^{\text{Coh}}} \mathbf{Coh}_{\sum_{i=1}^k \alpha_i}$$

where $\widetilde{\mathbf{Coh}}_{\underline{\alpha}}$ is the stack of flag of subsheaves $\mathcal{E}_1 \subseteq \mathcal{E}_2 \subseteq \dots \subseteq \dots \mathcal{E}_k$ with $\mathcal{E}_i$ of the class $\sum_{j=1}^i \alpha_j$.

Proposition 2.1. (1) The morphism $\text{q}_{\underline{\alpha}}^{\text{Coh}}$ is a vector bundle stack of relative dimension $-\sum_{i=1}^{k-1} \left\langle \sum_{j=1}^i \alpha_j, \alpha_{i+1} \right\rangle$.
(2) The morphism $\text{p}_{\underline{\alpha}}^{\text{Coh}}$ is schematic and proper.

Proof. To show (i), we can first consider when $\underline{\alpha} = (\alpha_1, \alpha_2)$, then for $\mathcal{E}_{\underline{\alpha}} = (\mathcal{E}_1, \mathcal{E}_2) \in \mathbf{Coh}_{\alpha_1} \times \mathbf{Coh}_{\alpha_2}$, the fiber $(\text{q}_{\underline{\alpha}}^{\text{Coh}})^{-1}(\mathcal{E}_{\underline{\alpha}})$ is the stack of extension $\text{Ext}^{\bullet}(\mathcal{E}_1, \mathcal{E}_2) = [\text{Ext}^1(\mathcal{E}_1, \mathcal{E}_2)/\text{Hom}(\mathcal{E}_1, \mathcal{E}_2)]$ and thus we get the desired property. In general for $(\mathcal{E}_1, \dots, \mathcal{E}_k) \in \mathbf{Coh}_{\underline{\alpha}}$, let $\underline{\alpha}' := (\alpha_1 + \alpha_2, \alpha_3, \dots, \alpha_k)$, we have the following diagram:

$$\begin{array}{ccc} (\text{q}_{\underline{\alpha}'}^{\text{Coh}})^{-1}(\mathcal{F}, \mathcal{E}_3, \dots, \mathcal{E}_k) & \longrightarrow & (\text{q}_{\underline{\alpha}}^{\text{Coh}})^{-1}(\mathcal{E}_1, \dots, \mathcal{E}_k) \\ \downarrow & & \downarrow \\ S & \xrightarrow{\mathcal{F}} & \text{Ext}^{\bullet}(\mathcal{E}_1, \mathcal{E}_2) \end{array}$$

Repeat the previous process we can reduce to the previous case of iterated extension stacks and get the result of (i). For (ii), we can consider the case $\underline{\alpha} = (\alpha_1, \alpha_2)$, then for $\mathcal{E} : S \rightarrow \mathbf{Coh}_{\alpha_1+\alpha_2}$, we get that its fiber product with $\text{p}_{\underline{\alpha}}^{\text{Coh}}$ is the proper Quot scheme $\text{Quot}_S(\mathcal{E}, \alpha)$. $\blacksquare$

Proposition 2.2. The functor Eis^{Coh} sends the subcategory of semi-simple objects to semi-simple ones.

Remark 2.3. As a consequence of 2.1, we have adjoint pairs:

$$\left(\text{q}_{\underline{\alpha}}^{\text{Coh}} \right)^! = \left(\text{q}_{\underline{\alpha}}^{\text{Coh}} \right)^* \dashv \left(\text{q}_{\underline{\alpha}}^{\text{Coh}} \right)_* \quad \left(\text{p}_{\underline{\alpha}}^{\text{Coh}} \right)_* = \left(\text{p}_{\underline{\alpha}}^{\text{Coh}} \right)_! \dashv \left(\text{p}_{\underline{\alpha}}^{\text{Coh}} \right)^!$$

and thus we have that Eis^{Coh} is left adjoint to CT^{Coh} .

For $\alpha, \beta \in \mathbf{Z}^{++}$, we have similar extension diagram $\mathbf{Bun}_{\alpha} \times \mathbf{Bun}_{\beta} \xleftarrow{\text{q}_{\alpha,\beta}^{\text{Bun}}} \mathbf{Bun}_{P_{\alpha,\beta}} \xrightarrow{\text{p}_{\alpha,\beta}^{\text{Bun}}} \mathbf{Bun}_{\alpha+\beta}$, where $\mathbf{Bun}_{P_{\alpha,\beta}}$ denotes the connected component of stack of parabolic torsors $\mathbf{Bun}_{P_{\text{rk}(\alpha), \text{rk}(\beta)}}$. We can define

Eisenstein series functor $\text{Eis}_{\alpha,\beta,!} = \left(\mathbf{p}_{\alpha,\beta}^{\text{Bun}}\right)_! \left(\mathbf{q}_{\alpha,\beta}^{\text{Bun}}\right)^*$ and constant term functor $\text{CT}_{\alpha,\beta,*} = \left(\mathbf{q}_{\alpha,\beta}^{\text{Bun}}\right)_* \left(\mathbf{p}_{\alpha,\beta}^{\text{Bun}}\right)_!$, we will abbreviate them as $\text{Eis}_!$ and CT_* if there is no ambiguity on the classes. In general, for any reductive group G and $P = MU$ parabolic subgroup in G with Levi subgroup M , we can define

$$\text{Eis}_{M,!}^G := (\mathbf{p}_P)_! \mathbf{q}_P^* \quad \dashv \quad \text{CT}_{G,*}^M := (\mathbf{q}_P)_* \mathbf{p}_P^!$$

where $\mathbf{Bun}_M \xleftarrow{\mathbf{q}_P} \mathbf{Bun}_P \xrightarrow{\mathbf{p}_P} \mathbf{Bun}_G$ is the parabolic induction diagram. Unlike Proposition 2.1, the morphism $\mathbf{p}_{\alpha,\beta}^{\text{Bun}}$ and $\mathbf{p}_P$ in general, is not proper, which means $\text{Eis}_!$ does not commute with Verdier duality. As a result, $\text{Eis}_!$ does not preserve perversity. To fix, we need to modify the Eisenstein series functor. The method is given by Braverman and Gaitsgory in [BG02], via so-called Drinfeld compactification.

For $\mathcal{Y}^{\circ} \subseteq \mathcal{Y}$ an open substack, set $\mathbf{Maps}_{\text{gen}}(X, \mathcal{Y} \supseteq \mathcal{Y}^{\circ})$ to be a subsheaf in groupoids of $\mathbf{Maps}(X, \mathcal{Y})$:

$$\mathbf{Maps}_{\text{gen}}(X, \mathcal{Y} \supseteq \mathcal{Y}^{\circ}) : \text{Sch}_{\mathbf{k}} \longrightarrow \text{Gpd};$$

$$S \mapsto \left\langle f \in \mathbf{Maps}(X, \mathcal{Y})(S) \left| \begin{array}{l} \text{for } \forall \bar{s} \rightarrow S \text{ geometric point, there exists} \\ U \subseteq X \times \bar{s} \text{ so that } f|_U \text{ factors through } \mathcal{Y}^{\circ} \end{array} \right. \right\rangle$$

Definition 2.4. Let Y be a scheme over $\mathbf{k}$, we say that Y is *strongly quasi-affine* if its global section is a finitely generated $\mathbf{k}$ -algebra and there is an open embedding $Y \hookrightarrow \text{Spec}(\Gamma(Y, \mathcal{O}_Y)) =: \bar{Y}$. Call $\bar{Y}$ the *affine closure* of Y .

Lemma 2.5. [BG02] Both G/U and $G/[P, P]$ are strongly quasi-affine.

We define the compactification of Bun_P to be

$$\begin{aligned} \widetilde{\text{Bun}}_P &:= \mathbf{Maps}_{\text{gen}}\left(X, G \backslash \overline{G/U} / M \supseteq \mathbf{BP}\right) \quad \text{Drinfeld's Compactification} \\ \overline{\text{Bun}}_P &:= \mathbf{Maps}_{\text{gen}}\left(X, G \backslash \overline{G/[P, P]} / (P/[P, P]) \supseteq \mathbf{BP}\right) \quad \text{Naive Compactification} \end{aligned}$$

We denote by $r_P : \widetilde{\text{Bun}}_P \longrightarrow \overline{\text{Bun}}_P$ induced by the projection $[P, P] = U \cdot [M, M] \twoheadrightarrow U$.

For $S \in \text{Sch}_{\mathbf{k}}$, we know that $\text{Bun}_P(S)$ is the groupoid of triples $(\mathcal{F}_G, \mathcal{F}_M, \kappa)$, where κ is a family of subbundles $\kappa^{\mathcal{V}} : (\mathcal{V}^{U(P)})_{\mathcal{F}_M} \hookrightarrow \mathcal{V}_{\mathcal{F}_G}$ associated with $\mathcal{V} \in \text{Rep } G$ (finite dimensional G -representations). And the *Plucker relation* (i.e. the following commutative diagram)

$$\begin{array}{ccc} (\mathcal{V}_1^U)_{\mathcal{F}_M} \otimes (\mathcal{V}_2^U)_{\mathcal{F}_M} & \xrightarrow{\kappa^{\mathcal{V}_1} \otimes \kappa^{\mathcal{V}_2}} & (\mathcal{V}_1 \otimes \mathcal{V}_2)_{\mathcal{F}_G} \\ \downarrow & & \downarrow \\ (\mathcal{V}_3^U)_{\mathcal{F}_M} & \xrightarrow{\kappa^{\mathcal{V}_3}} & (\mathcal{V}_3)_{\mathcal{F}_G} \end{array}$$

holds for every G -module map $\mathcal{V}_1 \otimes \mathcal{V}_2 \rightarrow \mathcal{V}_3$. In fact, by definition $\text{Bun}_P(S)$ classifies the data of

$$\left(\mathcal{F}_G, \mathcal{F}_M, \kappa : \mathcal{F}_G = \mathcal{F}_M \times^M G \longrightarrow \mathcal{F}_M \times^M G/U(P) \right)$$

We can thus easily see by Tannakian the equivalence of two definitions of Bun_P .

Now let H be either U or $[P, P]$, then $\mathbf{Maps}_{\text{gen}} \left(G \backslash \overline{G/H} / (P/H) \supseteq \mathbf{BP} \right) (S)$ is the groupoid of triples:

$$(\mathcal{F}_G, \mathcal{F}_{P/H}, \kappa)$$

where as above, κ is a family of maps of coherent sheaves

$$\kappa^{\mathcal{V}} : (\mathcal{V}^H)_{\mathcal{F}_{P/H}} \rightarrow \mathcal{V}_{\mathcal{F}_G}$$

so that the restriction of $\kappa^{\mathcal{V}}$ on $X \times \bar{s}$ is injective for any geometric point $\bar{s} \rightarrow S$.

Theorem 2.6. *The natural maps $\widetilde{\mathfrak{p}}_P : \widetilde{\text{Bun}}_P \rightarrow \text{Bun}_G$ and $\overline{\mathfrak{p}}_P : \overline{\text{Bun}}_P \rightarrow \text{Bun}_G$ are schematic and proper.*

Definition 2.7. We define the **compactified Eisenstein series functor** $\widetilde{\text{Eis}}_P^G : \text{Shv}(\text{Bun}_M) \longrightarrow \text{Shv}(\text{Bun}_G)$

$$\widetilde{\text{Eis}}_M^G : \mathcal{F} \mapsto (\widetilde{\mathfrak{p}}_P)_* \left(\mathbf{IC}_{\widetilde{\text{Bun}}_P} \otimes \widetilde{\mathfrak{q}}_P^*(\mathcal{F}) \otimes \left(\overline{\mathbb{Q}}_\ell(\frac{1}{2})[1] \right)^{-\otimes \dim(\text{Bun}_M)} \right)$$

Remark 2.8. Notice that when $P = B$ is Borel, we have $\widetilde{\text{Bun}}_B = \overline{\text{Bun}}_B$. We will call the associated Eisenstein series functor the *principal series* and denote it by $\overline{\text{Eis}}_T^G$. For convenience, we will abbreviate compactified Eisenstein series as $\widetilde{\text{Eis}}$ or $\overline{\text{Eis}}$ when there is no ambiguity in groups.

The notion of *principal series* goes also for $\text{Eis}_{\underline{\alpha}}^{\text{Coh}}$, we say it is principal series if $\text{rk}(\alpha_i) \leq 1$.

It is easy to see that $\text{Eis}_M^G \circ \text{Eis}_L^M = \text{Eis}_L^G$ for Levis $L \subseteq M \subseteq G$, however it is highly non-trivial that

Theorem 2.9. *For any Levis $L \subseteq M \subseteq G$, we have a canonical isomorphism*

$$\widetilde{\text{Eis}}_M^G \circ \widetilde{\text{Eis}}_L^M \cong \widetilde{\text{Eis}}_L^G$$

The Theorem is proved in [BG02] for $\widetilde{\text{Eis}} \circ \overline{\text{Eis}} \cong \widetilde{\text{Eis}}$. We attach the proof of general cases in the Appendix, which is a mimic of the same method in [BG02]. Similarly for the cases of $\mathbf{Coh}_{\alpha}$, we have:

Proposition 2.10. *The functor Eis^{Coh} is associative, namely for $\underline{\alpha} = (\alpha_1, \alpha_2, \alpha_3)$, we have*

$$\text{Eis}_{\alpha_1+\alpha_2, \alpha_3}^{\text{Coh}} \circ \left(\text{Eis}_{\alpha_1, \alpha_2}^{\text{Coh}} \otimes \text{Id}_{\alpha_3} \right) = \text{Eis}_{\alpha_1, \alpha_2+\alpha_3}^{\text{Coh}} \circ \left(\text{Id}_{\alpha_1} \otimes \text{Eis}_{\alpha_2, \alpha_3}^{\text{Coh}} \right) =: \text{Eis}_{\underline{\alpha}}^{\text{Coh}}$$

Similarly, the functor CT^{Coh} is coassociative:

$$\left(\text{CT}_{\alpha_1, \alpha_2}^{\text{Coh}} \otimes \text{Id}_{\alpha_3} \right) \circ \text{CT}_{\alpha_1 + \alpha_2, \alpha_3}^{\text{Coh}} = \left(\text{Id}_{\alpha_1} \otimes \text{CT}_{\alpha_2, \alpha_3}^{\text{Coh}} \right) \circ \text{CT}_{\alpha_1, \alpha_2 + \alpha_3}^{\text{Coh}} =: \text{CT}_{\alpha}^{\text{Coh}}$$

Proof. We can consider the following diagram:

$$\begin{array}{ccccc}
\text{Coh}_{\alpha_1} \times \text{Coh}_{\alpha_2} \times \text{Coh}_{\alpha_3} & \xleftarrow{q_{\alpha_1, \alpha_2}^{\text{Coh}} \times \text{id}_{\alpha_3}} & \widetilde{\text{Coh}}_{\alpha_1, \alpha_2} \times \text{Coh}_{\alpha_3} & \xrightarrow{p_{\alpha_1, \alpha_2}^{\text{Coh}} \times \text{id}_{\alpha_3}} & \text{Coh}_{\alpha_1 + \alpha_2} \times \text{Coh}_{\alpha_3} \\
\uparrow \text{id}_{\alpha_1} \times q_{\alpha_2, \alpha_3}^{\text{Coh}} & & & & \uparrow q_{\alpha_1 + \alpha_2, \alpha_3}^{\text{Coh}} \\
\text{Coh}_{\alpha_1} \times \widetilde{\text{Coh}}_{\alpha_2, \alpha_3} & & & & \widetilde{\text{Coh}}_{\alpha_1 + \alpha_2, \alpha_3} \\
\downarrow \text{id}_{\alpha_1} \times p_{\alpha_2, \alpha_3}^{\text{Coh}} & & & & \downarrow p_{\alpha_1 + \alpha_2, \alpha_3}^{\text{Coh}} \\
\text{Coh}_{\alpha_1} \times \text{Coh}_{\alpha_2 + \alpha_3} & \xleftarrow{q_{\alpha_1, \alpha_2 + \alpha_3}^{\text{Coh}}} & \widetilde{\text{Coh}}_{\alpha_1, \alpha_2 + \alpha_3} & \xrightarrow{p_{\alpha_1, \alpha_2 + \alpha_3}^{\text{Coh}}} & \text{Coh}_{\alpha_1 + \alpha_2 + \alpha_3}
\end{array}$$

along the upper-right circuit gives the left side of above equality, along the lower-left circuit gives the right side. Let $\tilde{\mathcal{E}}_{[\alpha_1, \alpha_2], \alpha_3}$ be the fiber product $\left(\widetilde{\text{Coh}}_{\alpha_1, \alpha_2} \times \text{Coh}_{\alpha_3} \right) \times_{\text{Coh}_{\alpha_1 + \alpha_2} \times \text{Coh}_{\alpha_3}} \widetilde{\text{Coh}}_{\alpha_1 + \alpha_2, \alpha_3}$ and $\tilde{\mathcal{E}}_{\alpha_1, [\alpha_2, \alpha_3]}$ be the fiber product $\left(\text{Coh}_{\alpha_1} \times \widetilde{\text{Coh}}_{\alpha_2, \alpha_3} \right) \times_{\text{Coh}_{\alpha_1} \times \text{Coh}_{\alpha_2 + \alpha_3}} \widetilde{\text{Coh}}_{\alpha_1, \alpha_2 + \alpha_3}$. It is not hard to see:

Lemma 2.11. *We have canonical isomorphism $\tilde{\mathcal{E}}_{[\alpha_1, \alpha_2], \alpha_3} \cong \tilde{\mathcal{E}}_{\alpha_1, [\alpha_2, \alpha_3]} \cong \widetilde{\text{Coh}}_{\alpha_1, \alpha_2 + \alpha_3}$.*

Then the associativity and coassociativity come can be deduced from the base change. ■

Proposition 2.12. *Let $\mathcal{C}$ and $\mathcal{D}$ be compactly generated categories, with dual categories $\mathcal{C}^\vee$ and $\mathcal{D}^\vee$. Assume there is adjoint pair $F : \mathcal{C} \xrightarrow{\perp} \mathcal{D} : G$ that induces $G^\vee : \mathcal{C}^\vee \xrightarrow{\perp} \mathcal{D}^\vee : F^\vee$, then we have*

$$\text{ev}_{\mathcal{D}}(F(\mathbf{c}) \otimes \mathbf{d}^\vee) = \text{ev}_{\mathcal{C}}(\mathbf{c} \otimes F^\vee(\mathbf{d}^\vee)) \quad \text{and} \quad \text{ev}_{\mathcal{D}}(G^\vee(\mathbf{c}^\vee) \otimes \mathbf{d}) = \text{ev}_{\mathcal{C}}(\mathbf{c}^\vee \otimes G(\mathbf{d}))$$

Proof. By definition and adjunction we have

$$\text{ev}_{\mathcal{C}}(\mathbf{c} \otimes F^\vee(\mathbf{d}^\vee)) = \text{Hom}_{\mathcal{C}^\vee}(\mathbf{c}^\vee, F^\vee(\mathbf{d}^\vee)) = \text{Hom}_{\mathcal{D}^\vee}(G^\vee((\mathbf{c}^\vee)^\vee), \mathbf{d}^\vee) = \text{ev}_{\mathcal{D}}\left(\left(G^\vee(\mathbf{c}^\vee)\right)^\vee \otimes \mathbf{d}^\vee\right)$$

So it remains to show that $(G^\vee(\mathbf{c}^\vee))^\vee = F(\mathbf{c})$, which is proved in [Gai15, Lemma 1.5.6]. ■

We have dual notions of functors Eis and CT . Notice that by Theorem 1.4, we have adjunction between compactly generated categories: $\text{Eis}_! \dashv \text{CT}_*$. Taking the dual categories and dual functors, we obtain:

$$\begin{aligned}
\text{Eis}_{\text{co},*} &:= (\text{CT}_*)^\vee : D(\mathbf{Bun}_M)_{\text{co}} \xrightarrow{\perp} D(\mathbf{Bun}_G)_{\text{co}} : (\text{Eis}_!)^\vee =: \text{CT}_{\text{co},?} \\
\text{Eis}_{\text{co},*}^{\text{Coh}} &:= (\text{CT}^{\text{Coh}})^\vee : D(\mathbf{Coh}_{\alpha,\beta})_{\text{co}} \xrightarrow{\perp} D(\mathbf{Coh}_{\alpha+\beta})_{\text{co}} : (\text{Eis}^{\text{Coh}})^\vee =: \text{CT}_{\text{co},*}^{\text{Coh}} \cong \text{CT}_{\text{co},?}^{\text{Coh}}
\end{aligned}$$

where we identify $D(\mathbf{Bun}_G)^\vee$ with $D(\mathbf{Bun}_G)_{\text{co}}$, as a corollary of Proposition 2.12, we have

Proposition 2.13. *For any reductive group G with pairing $\langle -, - \rangle_{D(\mathbf{Bun}_G)}$ and $\mathbf{P} \in D(\mathbf{Bun}_M)$, $\mathbf{Q} \in D(\mathbf{Bun}_G)_{\text{co}}$, we have*

$$\langle \text{Eis}!(\mathbf{P}), \mathbf{Q} \rangle_{D(\mathbf{Bun}_G)} = \langle \mathbf{P}, \text{CT}_{\text{co},?}(\mathbf{Q}) \rangle_{D(\mathbf{Bun}_M)}$$

Similarly, under the pairing on $D(\mathbf{Coh}_\alpha)$, for $\mathbf{P} \in D(\mathbf{Coh}_{\alpha,\beta})$, $\mathbf{Q} \in D(\mathbf{Coh}_{\alpha+\beta})_{\text{co}}$, we have

$$\langle \text{Eis}^{\text{Coh}}(\mathbf{P}), \mathbf{Q} \rangle_{D(\mathbf{Coh}_{\alpha+\beta})} = \langle \mathbf{P}, \text{CT}_{\text{co},*}^{\text{Coh}}(\mathbf{Q}) \rangle_{D(\mathbf{Coh}_{\alpha,\beta})}$$

2.1 The Langlands formula

The classical Langlands formula (see for example [MW95]) gives a re-interpretation of the constant term of Eisenstein series. Fix G be a reductive group, one can prove for automorphic forms on G and its Levis the following formula:

$$\text{CT}_G^M \circ \text{Eis}_L^G(f) = \sum_{w \in W_L \backslash W / W_M} \text{Eis}_{M^w}^M \circ \mathcal{R}_w \circ \text{CT}_L^{wL}(f)$$

where W_L is the sub-Weyl group of W Weyl group of G and w is a representative in the double coset. For each w , the groups M^w and wL are Levi subgroup of M and L respectively, such that they are conjugate by w . The middle term $\mathcal{R}_w$ on the right hand side is the induced conjugation on automorphic forms by w . For sheaves on $\mathbf{Bun}_G$ or $\mathbf{Coh}_\alpha$, we can also deduce a formula in this form. For tuple $\underline{r} = (r_1, \dots, r_s) \in \mathbb{Z}^s$, $\text{Eis}_{\underline{r}}^{\text{Coh}}$ denotes for $\bigoplus_{(r_i, d_i) \in \mathbb{Z}^+} \text{Eis}_{(r_1, d_1), \dots, (r_s, d_s)}^{\text{Coh}}$ and similarly $\text{CT}_{\underline{r}}^{\text{Coh}} = \bigoplus_{(r_i, d_i) \in \mathbb{Z}^+} \text{CT}_{(r_1, d_1), \dots, (r_s, d_s)}^{\text{Coh}}$. For two tuples $\underline{r} = (r_1, \dots, r_{s_1})$ and $\underline{r}' = (r'_1, \dots, r'_{s_2})$ with the same sum n , we say a matrix $\mathbb{M}$ of size $s \times s'$ is a *contingency matrix* from $\underline{r}$ to $\underline{r}'$ if its i -th row sums to r_i and its j -th column sums to r'_j .

Theorem 2.14 (Langlands formula for $\mathbf{Coh}$). *The composition of functor $\text{CT}_{r,s}^{\text{Coh}} \circ \text{Eis}_{p,q}^{\text{Coh}}$ admits a filtration with graded pieces are classified by contingency matrices from $[r, s]$ to $[p, q]$. In particular, let $\mathbb{M} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ be a contingency matrix from $[p, q]$ to $[r, s]$, the graded piece associated with $\mathbb{M}$ is given by*

$$\left(\text{Eis}_{a,c}^{\text{Coh}} \times \text{Eis}_{b,d}^{\text{Coh}} \right) \circ \mathcal{R}_b^c \circ \left(\text{CT}_{a,b}^{\text{Coh}} \times \text{CT}_{c,d}^{\text{Coh}} \right)$$

where $\mathcal{R}_b^c$ is induced equivalence by the swapping isomorphism $\mathbf{Coh}_{b,c} \xrightarrow{\cong} \mathbf{Coh}_{c,b} : (\mathcal{F}_1, \mathcal{F}_2) \mapsto (\mathcal{F}_2, \mathcal{F}_1)$.

Remark 2.15. To compare Theorem 2.14 with the classical Langlands formula above, we notice that $[p, q]$ and $[r, s]$ determines Levi subgroups $L = \text{GL}_p \times \text{GL}_q$ and $M = \text{GL}_r \times \text{GL}_s$ respectively. In this case, the set of contingency matrices from $[p, q]$ to $[r, s]$ are in bijection with the double coset $W_L \backslash W / W_M$.

Proof. We let $\widetilde{\mathcal{E}}_{p,q}^{r,s}$ denote the fiber product $\widetilde{\mathbf{Coh}}_{p,q} \times_{\mathbf{Coh}_n} \widetilde{\mathbf{Coh}}_{r,s}$. Then for any affine scheme S , $\widetilde{\mathcal{E}}_{p,q}^{r,s}(S)$ classifies the crosses as the following left diagram Figure 1:

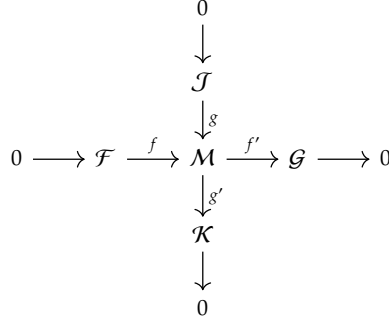


Figure 1: The cross diagram

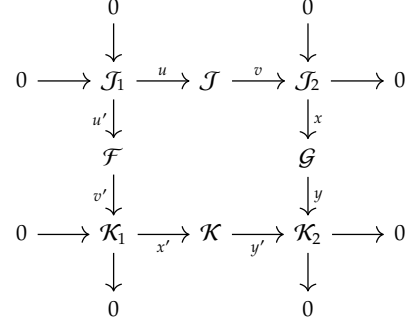


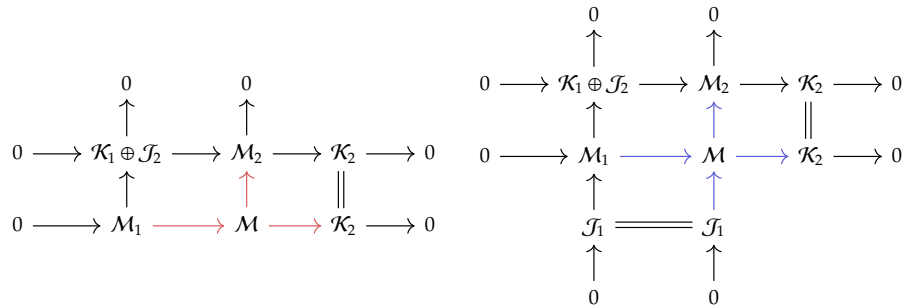
Figure 2: The square diagram

We fix $\mathcal{F}, \mathcal{G}, \mathcal{J}, \mathcal{K} \in \text{Coh}(X)$ with ranks p, q, r, s respectively. Let $\mathbf{Cr}_\mathcal{M}$ be the set of all choices (f, f', g, g') of the cross with middle term being $\mathcal{M}$. For any element in $\mathbf{Cr}_\mathcal{M}$, we can take

$$\mathcal{K}_1 := \text{Im}(g'f); \mathcal{K}_2 := \text{Coker}(g'f); \mathcal{J}_1 := \text{Ker}(f'g); \mathcal{J}_2 := \text{Im}(f'g).$$

We can deduce from the short exactness of both arrows in the cross that $\text{Ker}(g'f) \cong \text{Ker}(f'g)$ and $\text{Coker}(g'f) \cong \text{Coker}(f'g)$, and thus starting from the cross, we get the right above square diagram of short exact sequences (Figure 2). Let $\mathbf{Sq}(\mathcal{J}_1, \mathcal{J}_2, \mathcal{K}_1, \mathcal{K}_2)$ be the set of square diagrams with vertices $\mathcal{J}_1, \mathcal{J}_2, \mathcal{K}_1, \mathcal{K}_2$. We have constructed a map $\Phi : \coprod \mathbf{Cr}_\mathcal{M} \rightarrow \coprod \mathbf{Sq}(\mathcal{J}_1, \mathcal{J}_2, \mathcal{K}_1, \mathcal{K}_2)$. Moreover, if we let $a = \text{rk}(\mathcal{J}_1), b = \text{rk}(\mathcal{K}_1), c = \text{rk}(\mathcal{J}_2), d = \text{rk}(\mathcal{K}_2)$, we can associate a cross diagram with a contingency matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ from $[p, q]$ to $[r, s]$ by matching the rank of vertices of its image under Φ .

For a square diagram Sq in $\mathbf{Sq}(\mathcal{J}_1, \mathcal{J}_2, \mathcal{K}_1, \mathcal{K}_2)$, we can obtain $\mathcal{M}_1$: push out of $\mathcal{J}_1$ by u, u' and $\mathcal{M}_2$: the fiber product $\mathcal{G} \times_{\mathcal{K}_2} \mathcal{K}$ by y, y' . In particular, consider the direct sum of $\sigma_1 = \mathcal{K}_1 \xrightarrow{x'} \mathcal{K} \xrightarrow{y'} \mathcal{K}_2$ and $\sigma_2 = \mathcal{J}_2 \xrightarrow{x} \mathcal{G} \xrightarrow{y} \mathcal{K}_2$, we get an extension $\sigma_1 \oplus \sigma_2 \in \text{Ext}^1(\mathcal{K}_2, \mathcal{K}_1 \oplus \mathcal{J}_2)$, the corresponding middle term is exactly $\mathcal{M}_2$. Similarly, we can also obtain $\mathcal{J}_1 \rightarrow \mathcal{M}_1 \rightarrow \mathcal{K}_1 \oplus \mathcal{J}_2$. If a cross $\text{Cr} \in \mathbf{Cr}_\mathcal{M}$ maps to Sq , then we obtain natural maps $\mathcal{M}_1 \rightarrow \mathcal{M}$ and $\mathcal{M} \rightarrow \mathcal{M}_2$. Moreover, they fit canonically into the following right octahedral diagram. Also, the inverse image $\Phi^{-1}(\text{Sq}) \cap \mathbf{Cr}_\mathcal{M}$ are in bijection with the following left diagram, where the red arrows vary according to the choice of crosses in $\mathbf{Cr}_\mathcal{M}$.



For a contingency matrix $\mathbb{M} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ from $[p, q]$ to $[r, s]$, we let $\tilde{\mathcal{E}}_{\mathbb{M}}(S)$ be the groupoid that classifies the classes of square diagrams whose vertices have ranks correspond to the entries of $\mathbb{M}$. We claim the stack $\tilde{\mathcal{E}}_{p,q}^{r,s}$ can be decomposed into strata $\bigcup_{\mathbb{M}} \tilde{\mathcal{E}}_{\mathbb{M}}$ that is labelled by all contingency matrices from $[p, q]$ to $[r, s]$. Clearly, Φ induces a functor between the functors that $\tilde{\mathcal{E}}_{p,q}^{r,s}$ and $\bigcup_{\mathbb{M}} \tilde{\mathcal{E}}_{\mathbb{M}}$ represent. Thus, to prove the claim, we need to compare the scale of isomorphism classes and automorphism groups. We can calculate the scale of $\Phi^{-1}(\text{Sq}) \cap \mathbf{Cr}_{\mathcal{M}}$, fix a square Sq with vertices $(\mathcal{J}_1, \mathcal{J}_2, \mathcal{K}_1, \mathcal{K}_2)$. For this fixed square diagram Sq, the scale $|\Phi^{-1}(\text{Sq}) \cap \mathbf{Cr}_{\mathcal{M}}|$ counts the number of extension $\xi \in \text{Ext}^1(\mathcal{K}_2, \mathcal{M}_1)$ with middle term $\mathcal{M}$ and number of morphisms $\mathcal{M} \rightarrow \mathcal{M}_2$ so that the above right octahedral diagram commutes. To see this, we can apply $\mathcal{H}\text{om}$ to the exact sequence $0 \rightarrow \mathcal{J}_1 \rightarrow \mathcal{M}_1 \rightarrow \mathcal{K}_1 \oplus \mathcal{J}_2 \rightarrow 0$, combining with the property that the category $\text{Coh}(X_S)$ has cohomological dimension one, we get the following long exact sequence:

$$\begin{aligned} 0 \rightarrow \text{Hom}(\mathcal{K}_2, \mathcal{J}_1) \rightarrow \text{Hom}(\mathcal{K}_2, \mathcal{M}_1) \rightarrow \text{Hom}(\mathcal{K}_2, \mathcal{K}_1 \oplus \mathcal{J}_2) \rightarrow \text{Ext}^1(\mathcal{K}_2, \mathcal{J}_1) \rightarrow \\ \rightarrow \text{Ext}^1(\mathcal{K}_2, \mathcal{M}_1) \xrightarrow{\varphi} \text{Ext}^1(\mathcal{K}_2, \mathcal{K}_1 \oplus \mathcal{J}_2) \rightarrow 0 \end{aligned}$$

We can see once the extension $\xi \in \text{Ext}^1(\mathcal{K}_2, \mathcal{M}_1)$ is fixed, there exists a map $\mathcal{M} \rightarrow \mathcal{M}_2$ that fits into the octahedral diagram if and only if $\varphi(\xi) = \sigma_1 \oplus \sigma_2 \in \text{Ext}^1(\mathcal{K}_2, \mathcal{K}_1 \oplus \mathcal{J}_2)$. Thus, we get:

$$|\Phi^{-1}(\text{Sq}) \cap \mathbf{Cr}_{\mathcal{M}}| = \frac{|\text{Aut}(\mathcal{M})|}{|\text{Hom}(\mathcal{K}_2, \mathcal{M}_1)|} \cdot |\text{Hom}(\mathcal{K}_2, \mathcal{K}_1 \oplus \mathcal{J}_2)| \cdot |\varphi^{-1}(\sigma_1 \oplus \sigma_2)|.$$

This shows our claim.

We can turn to the composition of functors Eis^{Coh} and CT^{Coh} , we have

$$\text{CT}_{r,s}^{\text{Coh}} \circ \text{Eis}_{p,q}^{\text{Coh}} = (\mathbf{p}_{r,s}^{\text{Coh}})_! (\mathbf{q}_{r,s}^{\text{Coh}})^* (\mathbf{p}_{p,q}^{\text{Coh}})_! (\mathbf{q}_{p,q}^{\text{Coh}})^* \cong (\mathbf{p}_{r,s}^{\text{Coh}})_! (\pi_2)_! (\pi_1)^* (\mathbf{q}_{p,q}^{\text{Coh}})^*$$

where π_i is the projection of $\tilde{\mathcal{E}}_{p,q}^{r,s}$ along the following diagram

$$\begin{array}{ccccc} \mathbf{Coh}_p \times \mathbf{Coh}_q & \xleftarrow{\mathbf{q}_{p,q}^{\text{Coh}}} & \widetilde{\mathbf{Coh}}_{p,q} & \xrightarrow{\mathbf{p}_{p,q}^{\text{Coh}}} & \mathbf{Coh}_n \\ & & \uparrow \pi_1 & & \uparrow \mathbf{q}_{r,s}^{\text{Coh}} \\ & & \bigcup_{\mathbb{M}} \tilde{\mathcal{E}}_{\mathbb{M}} = \tilde{\mathcal{E}}_{p,q}^{r,s} & \xrightarrow{\pi_2} & \widetilde{\mathbf{Coh}}_{r,s} \\ & & & & \downarrow \mathbf{p}_{r,s}^{\text{Coh}} \\ & & & & \mathbf{Coh}_r \times \mathbf{Coh}_s \end{array}$$

For each contingency matrix $\mathbb{M}$, we can complete the above diagram to a big commutative diagram:

$$\begin{array}{ccccc}
\mathbf{Coh}_p \times \mathbf{Coh}_q & \xleftarrow{q_{p,q}^{\text{Coh}}} & \widetilde{\mathbf{Coh}}_{p,q} & \xrightarrow{p_{p,q}^{\text{Coh}}} & \mathbf{Coh}_n \\
\uparrow p_{a,b}^{\text{Coh}} \times p_{c,d}^{\text{Coh}} & & \uparrow \pi_1^{\mathbb{M}} & & \uparrow q_{r,s}^{\text{Coh}} \\
\widetilde{\mathbf{Coh}}_{a,b} \times \widetilde{\mathbf{Coh}}_{c,d} & \xleftarrow{\tilde{\pi}_1^{\mathbb{M}}} & \tilde{\mathcal{E}}_{\mathbb{M}} & \xrightarrow{\pi_2^{\mathbb{M}}} & \widetilde{\mathbf{Coh}}_{r,s} \\
\downarrow q_{a,b}^{\text{Coh}} \times q_{c,d}^{\text{Coh}} & & \downarrow \tilde{\pi}_2^{\mathbb{M}} & & \downarrow p_{r,s}^{\text{Coh}} \\
\mathbf{Coh}_{a,b,c,d} \cong \mathbf{Coh}_{a,c,b,d} & \xleftarrow{q_{a,c}^{\text{Coh}} \times q_{b,d}^{\text{Coh}}} & \widetilde{\mathbf{Coh}}_{a,c} \times \widetilde{\mathbf{Coh}}_{b,d} & \xrightarrow{p_{a,c}^{\text{Coh}} \times p_{b,d}^{\text{Coh}}} & \mathbf{Coh}_r \times \mathbf{Coh}_s
\end{array}$$

where $\pi_i^{\mathbb{M}}$ is the composition $\tilde{\mathcal{E}}_{\mathbb{M}} \xrightarrow{\iota_{\mathbb{M}}} \tilde{\mathcal{E}}_{p,q}^{r,s}$ with π_i . Note that by construction, $\tilde{\mathcal{E}}_{\mathbb{M}}$ is the fiber product $(\widetilde{\mathbf{Coh}}_{a,b} \times \widetilde{\mathbf{Coh}}_{c,d}) \times_{\mathbf{Coh}_{a,b,c,d}} (\widetilde{\mathbf{Coh}}_{a,c} \times \widetilde{\mathbf{Coh}}_{b,d})$, where the morphism $\widetilde{\mathbf{Coh}}_{a,c} \times \widetilde{\mathbf{Coh}}_{b,d} \rightarrow \mathbf{Coh}_{a,b,c,d}$ is given by the composition

$$\widetilde{\mathbf{Coh}}_{a,c} \times \widetilde{\mathbf{Coh}}_{b,d} \xrightarrow{q_{a,c}^{\text{Coh}} \times q_{b,d}^{\text{Coh}}} \mathbf{Coh}_{a,b,c,d} \xrightarrow{\text{Id}_a \times \tau_b^c \times \text{Id}_d} \mathbf{Coh}_{a,b,c,d}$$

where τ_b^c is the swapping isomorphism.

For each contingency matrix $\mathbb{M}$, we can consider the following functor:

$$\begin{aligned}
(\text{Eis}_{a,c}^{\text{Coh}} \times \text{Eis}_{b,d}^{\text{Coh}}) \circ \mathcal{R}_b^c \circ (\text{CT}_{a,b}^{\text{Coh}} \times \text{CT}_{c,d}^{\text{Coh}}) &= (p_{a,c}^{\text{Coh}} \times p_{b,d}^{\text{Coh}})_! (q_{a,c}^{\text{Coh}} \times q_{b,d}^{\text{Coh}})^* \circ \mathcal{R}_b^c \circ (p_{a,b}^{\text{Coh}} \times p_{c,d}^{\text{Coh}})_! (q_{a,b}^{\text{Coh}} \times q_{c,d}^{\text{Coh}})^* \\
&\cong (p_{a,c}^{\text{Coh}} \times p_{b,d}^{\text{Coh}})_! (\tilde{\pi}_2^{\mathbb{M}})_! (\tilde{\pi}_1^{\mathbb{M}})^* (q_{a,b}^{\text{Coh}} \times q_{c,d}^{\text{Coh}})^* \\
&\cong (p_{r,s}^{\text{Coh}})_! (\pi_2^{\mathbb{M}})_! (\pi_1^{\mathbb{M}})^* (q_{p,q}^{\text{Coh}})^*
\end{aligned}$$

On the other hand, by adjointness we have $(\iota_{\mathbb{M}})_! (\iota_{\mathbb{M}})^! \rightarrow \text{Id}$ and thus we get

$$\begin{aligned}
(p_{r,s}^{\text{Coh}})_! (\pi_2^{\mathbb{M}})_! (\pi_1^{\mathbb{M}})^* (q_{p,q}^{\text{Coh}})^* &\cong (p_{r,s}^{\text{Coh}})_! (\pi_2)_! (\iota_{\mathbb{M}})_! (\iota_{\mathbb{M}})^! (\pi_1)^* (q_{p,q}^{\text{Coh}})^* \\
&\longrightarrow (p_{r,s}^{\text{Coh}})_! (\pi_2)_! (\pi_1)^* (q_{p,q}^{\text{Coh}})^* \cong (p_{r,s}^{\text{Coh}})_! (q_{r,s}^{\text{Coh}})^* (p_{p,q}^{\text{Coh}})_! (q_{p,q}^{\text{Coh}})^* = \text{CT}_{r,s}^{\text{Coh}} \circ \text{Eis}_{p,q}^{\text{Coh}}
\end{aligned}$$

2.2 Proof of the miraculosity

The following properties can be easily deduced:

$$\begin{aligned}
\text{Eis}_{\text{co},*}^{\text{Coh}} \circ (j_{\underline{\alpha}})_{\text{co},*} &\cong (j_{\underline{\alpha}})_{\text{co},*} \circ (j_{\underline{\alpha}})^* \circ \text{Eis}_{*,*}^{\text{coh}} \circ (j_{\underline{\alpha}})_* \\
\text{CT}_{\underline{\alpha},\text{co},*}^{\text{Coh}} \circ (j_{\underline{\alpha}})_{\text{co},*} &\cong (j_{\underline{\alpha}})_{\text{co},*} \circ (j_{\underline{\alpha}})^* \circ \text{CT}_{\underline{\alpha}}^{\text{coh}} \circ (j_{\underline{\alpha}})_*
\end{aligned}$$

Moreover, we have the following important property, which is parallel to [DG16, Proposition B.2.2]

Proposition 2.16. *Take the complement of very unstable locus in $\mathbf{Coh}_{\alpha}$, we get an open substack $j_{\alpha}^{\text{nvu}} : \mathbf{Coh}_{\alpha}^{\text{nvu}} \hookrightarrow$*

$\mathbf{Coh}_\alpha$. Then, for any $\mathbf{F} \in D(\mathbf{Coh}_\alpha)_{\text{cusp}}$, there are isomorphisms

$$(j_\alpha^{\text{nvu}})_! \circ (j_\alpha^{\text{nvu}})^*(\mathbf{F}) \xrightarrow{\cong} \mathbf{F} \xrightarrow{\cong} (j_\alpha^{\text{nvu}})_* \circ (j_\alpha^{\text{nvu}})^*(\mathbf{F})$$

Proof. Let $\iota^{\text{vu}} : \mathbf{Coh}_\alpha^{\text{vu}} \hookrightarrow \mathbf{Coh}_\alpha$ be the locally closed embedding of very unstable locus. We want to show $(\iota^{\text{vu}})^!(\mathbf{F}) = (\iota^{\text{vu}})^*(\mathbf{F}) = 0$ for any $\mathbf{F} \in D(\mathbf{Coh}_\alpha)_{\text{cusp}}$. This follows easily from the following Lemma, which is a slight modification of [DG16, Lemma B.3.2]:

Lemma 2.17. *Let $\underline{\beta}$ be a very unstable HN type of α , then the induction diagram can be restricted to*

$$\mathbf{Coh}_{\underline{\beta}}^{\text{ss}} \xleftarrow{q_{\underline{\beta}}^{\text{Coh}}} \widetilde{\mathbf{Coh}}_{\underline{\beta}} \xrightarrow{p_{\underline{\beta}}^{\text{Coh}}} \text{HN}^{-1}(\underline{\beta})$$

In other words, the preimage of $\mathbf{Coh}_{\underline{\beta}}^{\text{ss}}$ under $q_{\underline{\beta}}^{\text{Coh}}$ in $\widetilde{\mathbf{Coh}}_{\underline{\beta}}$ is included in the preimage of $\text{HN}^{-1}(\underline{\beta})$ under $p_{\underline{\beta}}^{\text{Coh}}$.

Moreover, $p_{\underline{\beta}}^{\text{Coh}}$ restrict to $\left(q_{\underline{\beta}}^{\text{Coh}}\right)^{-1}(\mathbf{Coh}_{\underline{\beta}}^{\text{ss}})$ is surjective onto $\text{HN}^{-1}(\underline{\beta})$.

Besides, because the filtration splits on $\mathbf{Coh}_{\underline{\beta}}^{\text{ss}}$, we can deduce that $q_{\underline{\beta}}^{\text{Coh}}$ over $\mathbf{Coh}_{\underline{\beta}}^{\text{ss}}$ is a unipotent gerbe, as it induces a bijection on the isomorphism classes. We consider the constant term

$$\text{CT}_{\underline{\beta}}^{\text{Coh}}|_{\text{HN}^{-1}(\underline{\beta})} \circ \left(\iota^{\text{vu}}|_{\text{HN}^{-1}(\underline{\beta})}\right)^!(\mathcal{F}) = \left(q_{\underline{\beta}}^{\text{Coh}}\right)_* \circ \left(p_{\underline{\beta}}^{\text{Coh}}\right)^! \circ \left(\iota^{\text{vu}}|_{\text{HN}^{-1}(\underline{\beta})}\right)^!(\mathcal{F})$$

Since over $\mathbf{Coh}_{\underline{\beta}}^{\text{ss}}$, $q_{\underline{\beta}}^{\text{Coh}}$ is a unipotent gerbe, it induces an equivalence $\left(q_{\underline{\beta}}^{\text{Coh}}\right)_*$ on this part, also $\left(p_{\underline{\beta}}^{\text{Coh}}\right)^!$ is conservative over $D\left(\text{HN}^{-1}(\underline{\beta})\right)$. By assumption, $\mathbf{F} \in D(\mathbf{Coh}_\alpha)_{\text{cusp}}$, thus $\text{CT}_{\underline{\beta}}^{\text{Coh}}(\mathbf{F}) = 0$. We can conclude that $\left(\iota^{\text{vu}}|_{\text{HN}^{-1}(\underline{\beta})}\right)^!(\mathbf{F}) = 0$ since $\text{CT}_{\underline{\beta}}^{\text{Coh}}|_{\text{HN}^{-1}(\underline{\beta})}$ is conservative.

Next, to show $\left(\iota^{\text{vu}}|_{\text{HN}^{-1}(\underline{\beta})}\right)^*(\mathbf{F}) = 0$, it is enough to show $\text{Hom}\left(\mathbf{F}, \left(\iota^{\text{vu}}|_{\text{HN}^{-1}(\underline{\beta})}\right)_*(\mathbf{F}')\right) = 0$. By above Lemma, it is enough to show $\text{Hom}\left(\mathbf{F}, \text{Eis}^{\text{Coh}}(\mathbf{E}')\right) = 0$ for all $\mathbf{E}' \in D\left(\mathbf{Coh}_{\underline{\beta}}^{\text{ss}}\right)$, since we have

$$\text{CT}^{\text{Coh}, -} \dashv \text{Eis}^{\text{Coh}} \dashv \text{CT}^{\text{Coh}}$$

The assertion follows again from the fact that $\mathbf{F}$ is cuspidal. ■

Corollary 2.18. *As a corollary of Proposition 2.16, we can show Ps-Id_* establishes an equivalence $D(\mathbf{Coh}_\alpha)_{\text{co, cusp}} \xrightarrow{\cong} D(\mathbf{Coh}_\alpha)_{\text{cusp}}$, whose inverse is given by $\mathbf{F} \mapsto (j_\alpha^{\text{nvu}})_{\text{co},*}(j_\alpha^{\text{nvu}})^*(\mathbf{F})$.*

Next, we would like to compare Ps-Id_* and $\text{Ps-Id}_!$ on the cuspidal parts. Let $\text{Ps-Id}_{\text{diff}}$ denote the cone of $\text{Ps-Id}_! \rightarrow \text{Ps-Id}_*[\text{shift}]$, then we have the following key proposition:

Proposition 2.19. *There is a filtration on $\text{Ps-Id}_{\text{diff}}$ whose graded pieces are of the form:*

$$\text{Eis}^{\text{Coh}} \circ \mathcal{R} \circ \text{Ps-Id}_* \circ \text{CT}_{\text{co},*}^{\text{Coh}}$$

proof to be added. As a conclusion, we can deduce that $\text{Ps-Id}_{\text{diff}}$ vanishes over co-cuspidal part. In other words, $\text{Ps-Id}_{\text{!}}$ and Ps-Id_* are equivalent up to a shift on $D(\mathbf{Coh})_{\text{co, cusp}}$.

3 Lusztig perverse sheaves

3.1 Definition and basic properties

We have seen that for each $\alpha \in \mathbf{Z}^+$, the stack $\mathbf{Coh}_\alpha$ can be exhibited by smooth quasi-compact open substacks $\mathbf{Coh}_\alpha^{>\mathcal{L}} \cong [\mathbf{Q}_\alpha^\mathcal{L}/G_\alpha]$. When X is over $\bar{\mathbb{F}}_q$, we denote by $\mathbf{1}_\alpha$ the colimit of perverse sheaves $\overline{\mathbb{Q}}_\ell[\mathbf{Q}_\alpha^\mathcal{L}/G_\alpha][\dim(\mathbf{Coh}_\alpha)]$, which lies in $\text{Perv}(\mathbf{Coh}_\alpha) \subseteq \text{Shv}_{\text{constr}}(\mathbf{Coh}_\alpha)$. Correspondingly, for $X/\mathbb{C}$, $\mathbf{1}_\alpha$ is the colimit of $\mathbb{C}[\mathbf{Q}_\alpha^\mathcal{L}/G_\alpha][\dim(\mathbf{Coh}_\alpha)]$, lies in $\text{Perv}(\mathbf{Coh}_\alpha) = \mathcal{D}\text{-mod}_{\text{reg.hol}}(\mathbf{Coh}_\alpha)^\vee \subseteq \mathcal{D}\text{-mod}(\mathbf{Coh}_\alpha)$.

Definition 3.1. By a **Lusztig perverse sheaf** on $\mathbf{Coh}_\alpha$ we mean a perverse sheaf whose shift is a simple subobject of Laumon's *Eisenstein sheaves* $\text{Eis}_\alpha^{\text{Coh}}(\mathbf{1}_{\alpha_1} \boxtimes \dots \boxtimes \mathbf{1}_{\alpha_k})$, where $\sum_{i=1}^k \alpha_i = \alpha$ and $\text{rk}(\alpha_i) \leq 1$. We let $\mathcal{P}_\alpha^{\text{Coh}}$ be the set of Lusztig perverse sheaves on $\mathbf{Coh}_\alpha$ and write $\mathcal{P}_r^{\text{Coh}} = \bigsqcup_{\text{rk } \alpha = r} \mathcal{P}_\alpha^{\text{Coh}}$. When $r = 0$, we also use $\mathcal{P}^{\text{Tor}}$ to denote for $\mathcal{P}_0^{\text{Coh}}$. Similarly, when $\text{rk}(\alpha) \geq 1$, a **Lusztig perverse sheaf** on $\mathbf{Bun}_\alpha$ is a perverse sheaf whose shift is a simple subobject of constructible sheaf/ $\mathcal{D}$ -module of the form $\overline{\text{Eis}}_\alpha(\mathbf{1}_{\alpha_1} \boxtimes \dots \boxtimes \mathbf{1}_{\alpha_k})$, where $\sum_{i=1}^l \alpha_i = \alpha$ and $\text{rk}(\alpha_i) = 1$. We denote by $\mathcal{P}_\alpha^{\text{Bun}}$ the set of Lusztig perverse sheaves on $\mathbf{Bun}_\alpha$ and write $\mathcal{P}_r^{\text{Bun}} = \bigsqcup_{\text{rk } \alpha = r} \mathcal{P}_\alpha^{\text{Bun}}$.

Definition 3.2. For $\alpha \in \mathbf{Z}^+$, we denote by $\mathcal{Q}_\alpha^{\text{Coh}}$ the category additively generated by $\mathcal{P}_\alpha^{\text{Coh}}$. We use the same notation for the DG-category which is the Karoubi completion of $\mathcal{Q}_\alpha^{\text{Coh}}$ and call it **Laumon-Lusztig category**, notice that the Karoubi completion is included in $\text{Ind}(\mathcal{Q}_\alpha^{\text{Coh}}) \subseteq \text{Shv}(\mathbf{Coh}_\alpha)$.

Similarly, for $\alpha \in \mathbf{Z}^{++}$, we denote by $\mathcal{Q}_\alpha^{\text{Bun}}$ the DG-category which is the (Karoubi completion) of the category additively generated by $\mathcal{P}_\alpha^{\text{Bun}}$.

Lemma 3.3. $\text{Eis}_{\alpha,\beta}^{\text{Coh}}$ sends $\mathcal{Q}_\alpha^{\text{Coh}} \times \mathcal{Q}_\beta^{\text{Coh}}$ to $\mathcal{Q}_{\alpha+\beta}^{\text{Coh}}$ and $\widetilde{\text{Eis}}_{\alpha,\beta}$ sends $\mathcal{Q}_\alpha^{\text{Bun}} \times \mathcal{Q}_\beta^{\text{Bun}}$ to $\mathcal{Q}_{\alpha+\beta}^{\text{Bun}}$.

Proof. The first half follows easily from the associativity of Eis^{Coh} and the properness of $\mathbf{q}^{\text{Coh}}$. The second half is, however more difficult, we have $\widetilde{\text{Eis}}(\overline{\text{Eis}} \boxtimes \overline{\text{Eis}}) = \overline{\text{Eis}}$ cf [BG02, Theorem 2.3.10] and Theorem 2.9 and we can combine it with the fact that $\widetilde{\mathbf{Bun}}_p \rightarrow \mathbf{Bun}_G$ is proper. $\blacksquare$

Proposition 3.4. The compactified principal Eisenstein series $\overline{\text{Eis}}$ for $\text{GL}_r(r \geq 1)$ is compatible with Eis^{Coh} . More concretely, for any $\underline{\alpha} = (\alpha_1, \dots, \alpha_r)$ with $\text{rk}(\alpha_i) = 1$, we have

$$J_\alpha^* \circ \text{Eis}_{\underline{\alpha}}^{\text{Coh}} \circ (J_\alpha)_! \cong \overline{\text{Eis}}_{\underline{\alpha}} \iff \text{Eis}_{\underline{\alpha}}^{\text{Coh}} \circ (J_\alpha)_! \cong (J_\alpha)_! \circ \overline{\text{Eis}}_{\underline{\alpha}}$$

Theorem 3.5. The set of Lusztig perverse sheaves $\mathcal{P}_r^{\text{Coh}}$ on $\mathbf{Coh}_r$ is in bijection with $\mathcal{P}_r^{\text{Bun}} \times \mathcal{P}^{\text{Tor}}$.

Proof. Begin with $\mathbf{P} \boxtimes \mathbf{T} \in \mathcal{P}_r^{\text{Bun}} \times \mathcal{P}^{\text{Tor}}$, we can consider the following diagram:

$$\begin{array}{ccccc} \mathbf{Bun}_\alpha \times \mathbf{Coh}_{\tau l} & \xleftarrow{\quad} & \mathcal{Y} & \xhookrightarrow{\iota^l} & \mathbf{Coh}_{\alpha+\tau l}^{\text{tor.deg} \leq l} \\ \downarrow J_\alpha \times \text{Id} & & \downarrow \cap & & \downarrow J_{\alpha+\tau l}^{\leq l} \\ \mathbf{Coh}_\alpha \times \mathbf{Coh}_{\tau l} & \xleftarrow{q^{\text{Coh}}} & \widetilde{\mathbf{Coh}}_{\alpha,\tau l} & \xrightarrow{p^{\text{Coh}}} & \mathbf{Coh}_{\alpha+\tau l} \end{array}$$

where $\mathcal{Y}$ is the fiber product $(\mathbf{Bun}_\alpha \times \mathbf{Coh}_{\tau l})_{\mathbf{Coh}_\alpha \times \mathbf{Coh}_{\tau l}} \times \widetilde{\mathbf{Coh}}_{\alpha, \tau l}$.

Lemma 3.6. *We have $\mathcal{Y} \cong \mathbf{Coh}^{\text{tor.deg}=l}$, and the right square in the above diagram commutes.*

We send $\mathbf{P} \boxtimes \mathbf{T}$ to

$$(\mathbf{P} \boxtimes \mathbf{T})^{\text{Coh}} := \text{Eis}^{\text{Coh}}((j_\alpha \times \text{Id})_{!*}(\mathbf{P} \boxtimes \mathbf{T})) \in \text{Shv}(\mathbf{Bun}_{\alpha+\tau l})$$

and we would like to see this is a simple perverse sheaf on $\mathbf{Coh}_{\alpha+\tau l}$. As Eis^{Coh} preserves perversity and we take the intermediate extension along open immersion $\mathbf{Bun}_\alpha \hookrightarrow \mathbf{Coh}_\alpha$, we can see immediately that $(\mathbf{P} \boxtimes \mathbf{T})^{\text{Coh}}$ is a perverse sheaf. Note that $\mathcal{Y} \rightarrow \mathbf{Bun}_\alpha \times \mathbf{Coh}_{\tau l}$ is a unipotent gerbe, then by smooth and proper base change, we have $(\mathbf{P} \boxtimes \mathbf{T})^{\text{Coh}} \cong \left(j_{\alpha+\tau l}^{\leq l}\right)_{!*}(\iota^l)_!(\mathbf{P} \boxtimes \mathbf{T})[\text{shift}]$. By smoothness and the fact that $\mathbf{Coh}_{\alpha+\tau l}^{\text{tor.deg} \leq l}$ is open in $\mathbf{Coh}_{\alpha+\tau l}$, it suffices to check that $(\iota^l)_!(\mathbf{P} \boxtimes \mathbf{T})$ is irreducible. But ι^l is proper, by decomposition theorem, it suffices to check the irreducibility of $\mathbf{P} \boxtimes \mathbf{T}$, which is automatic. We then verify that $(\mathbf{P} \boxtimes \mathbf{T})^{\text{Coh}}$ is in principal series, i.e. a component of some $\text{Eis}^{\text{Coh}}(\mathbf{1}_{\alpha_1} \boxtimes \dots \boxtimes \mathbf{1}_{\alpha_k})$. By assumption, $\mathbf{P} \in \mathcal{P}_r^{\text{Bun}}$, then there exists $\underline{\alpha}' = (\alpha'_1, \dots, \alpha'_m)$ such that $\mathbf{P} \subseteq \overline{\text{Eis}}(\mathbf{1}_{\alpha'_1} \boxtimes \dots \boxtimes \mathbf{1}_{\alpha'_m})$. By compatibility between $\overline{\text{Eis}}$ and Eis^{Coh} , we have

$$\begin{aligned} \text{Eis}_{m, \tau l}^{\text{Coh}}\left((j_m)_{!*}\left(\overline{\text{Eis}}(\mathbf{1}_{\alpha'_1} \boxtimes \dots \boxtimes \mathbf{1}_{\alpha'_m})\right) \boxtimes \mathbf{T}\right) &= \text{Eis}_{m, \tau l}^{\text{Coh}}\left(\text{Eis}^{\text{Coh}}\left((j_{\alpha_1})_{!*}(\mathbf{1}_{\alpha'_1}) \boxtimes \dots \boxtimes (j_{\alpha_1})_{!*}(\mathbf{1}_{\alpha'_m})\right) \boxtimes \mathbf{T}\right) \\ &= \text{Eis}^{\text{Coh}}\left(\mathbf{1}_{\alpha'_1} \boxtimes \dots \boxtimes \mathbf{1}_{\alpha'_m} \boxtimes \mathbf{T}\right) \end{aligned}$$

We can see $(\mathbf{P} \boxtimes \mathbf{T})^{\text{Coh}} \in \mathcal{P}_\alpha^{\text{Coh}}$ by definition.

Conversely, let $\mathbf{P} \in \mathcal{P}_r^{\text{Coh}}$, then there exists a minimal integer $d \gg 0$ such that $\text{Supp}(\mathbf{P}) \cap \mathbf{Coh}^{\text{tor.deg} \leq d} \neq \emptyset$. We claim that there exists $\tilde{\mathbf{P}}$ such that $(\iota^d)_!(\tilde{\mathbf{P}}) \cong (j_\alpha^{\leq d})^*(\mathbf{P}) \in \text{Perv}(\mathbf{Coh}^{\text{tor.deg} \leq d})$. In fact, since $\mathbf{P}$ is simple, its restriction to open locus $(j_\alpha^{\leq d})^*(\mathbf{P})$ is also simple, so it must be in the form of $\mathbf{IC}_Z(\mathcal{L})$, where $Z \subseteq \mathbf{Coh}^{\text{tor.deg} \leq d}$ locally closed. There exists a minimal $0 \leq d' \leq d$ such that $Z \cap \mathbf{Coh}^{\text{tor.deg} = d'} \neq \emptyset$. If $d' < d$, then it contradicts to our assumption on d , this forces $d' = d$, but this means Z is included in $\mathbf{Coh}^{\text{tor.deg} = d}$, then we find the $\tilde{\mathbf{P}}$ we want, which is simple. Also use the fact that $\pi^{\text{tor.deg} = d} : \mathbf{Coh}_\alpha^{\text{tor.deg} = d} \rightarrow \mathbf{Bun}_{\alpha-\tau d} \times \mathbf{Coh}_{\tau d}$ is a unipotent gerbe, we can deduce that $(\pi^{\text{tor.deg} = d})_![\text{shift}]$ induces an equivalence $\text{Perv}(\mathbf{Coh}_\alpha^{\text{tor.deg} = d}) \cong \text{Perv}(\mathbf{Bun}_{\alpha-\tau d}) \times \text{Perv}(\mathbf{Coh}_{\tau d})$, so we obtain from $\tilde{\mathbf{P}}$ a pair of simple perverse sheaves $(\mathbf{P}', \mathbf{T}) \in \text{Perv}(\mathbf{Bun}_{\alpha-\tau d}) \times \text{Perv}(\mathbf{Coh}_{\tau d})$. Moreover, since $\mathbf{P} \cong (j_\alpha^{\leq d})_{!*}(j_\alpha^{\leq d})^*(\mathbf{P})$, we have $(\mathbf{P}' \boxtimes \mathbf{T})^{\text{Coh}} \cong \mathbf{P}$. It remains to show that $\mathbf{P}'$ and $\mathbf{T}$ are in principal series.

Lemma 3.7. *For any $\mathbf{P}_1 \boxtimes \mathbf{T}_1 \in \mathbf{Coh}_\beta \times \mathbf{Coh}_\gamma$, we have $\mathbf{P}_1 \boxtimes \mathbf{T}_1 \subseteq \text{CT}_{\beta, \gamma}^{\text{Coh}} \circ \text{Eis}_{\beta, \gamma}^{\text{Coh}}(\mathbf{P}_1 \boxtimes \mathbf{T}_1)$.*

Proof. This is an easy consequence of Langlands formula (Theorem 2.14). ■

Apply to $(j_{\alpha-\tau d})_{!*}(\mathbf{P}') \boxtimes \mathbf{T}$, we only need to show $\text{CT}_{\alpha-\tau d, \tau d}^{\text{Coh}}(\mathbf{P})$ is in principal series.

Lemma 3.8. For $\underline{\alpha} = (\alpha_1, \dots, \alpha_r)$ and $\beta, \gamma \in \mathbf{Z}^+$ such that $\sum \alpha_i = \beta + \gamma$, then

$$\mathrm{CT}_{\beta, \gamma}^{\mathrm{Coh}} \left(\mathrm{Eis}^{\mathrm{Coh}} (\mathbf{1}_{\alpha_1} \boxtimes \dots \boxtimes \mathbf{1}_{\alpha_r}) \right) = \bigoplus_{\underline{\beta}, \underline{\gamma}} \mathrm{Eis}^{\mathrm{Coh}} (\mathbf{1}_{\beta_1} \boxtimes \dots \boxtimes \mathbf{1}_{\beta_r}) \boxtimes \mathrm{Eis}^{\mathrm{Coh}} (\mathbf{1}_{\gamma_1} \boxtimes \dots \boxtimes \mathbf{1}_{\gamma_r})$$

where $\underline{\beta}, \underline{\gamma}$ runs over $(\beta_1, \dots, \beta_r) \in (\mathbf{Z}^+)^r$ and $(\gamma_1, \dots, \gamma_r) \in (\mathbf{Z}^+)^r$ such that $\sum \beta_i = \beta, \sum \gamma_i = \gamma$.

Proof of Lemma. The graded pieces of the above left are given by Langlands formula 2.14. It remains to show that the extensions among them are trivial. We have the following Lemma of Lusztig:

Lemma 3.9. For $f : Y \rightarrow Z$ a morphism between varieties such that there exists a finite stratification $Y = \bigsqcup_{\sigma} Y_{\sigma}$, restriction of f on Y_{σ} can be decomposed into $Y_{\sigma} \xrightarrow{f_{\sigma}} \tilde{Y}_{\sigma} \xrightarrow{g_{\sigma}} Z$ and that f_{σ} is vector bundle, g_{σ} is proper. Then we have $f_! \left(\overline{\mathbb{Q}}_{\ell} \right) = \bigoplus_{\sigma} (f|_{Y_{\sigma}})_! \left(\overline{\mathbb{Q}}_{\ell} \right)$.

We can apply Lemma 3.9 to the restriction of $\mathbf{q}^{\mathrm{Coh}}$ onto each $\mathbf{Coh}^{>\mathcal{L}}$. ■

By our assumption, $\mathbf{P}$ is in principal series, thus $\mathrm{CT}_{\alpha-\tau d, \tau d}^{\mathrm{Coh}}(\mathbf{P}) \subseteq \mathrm{CT}_{\alpha-\tau d, \tau d}^{\mathrm{Coh}} \left(\mathrm{Eis}^{\mathrm{Coh}} (\mathbf{1}_{\alpha_1} \boxtimes \dots \boxtimes \mathbf{1}_{\alpha_r}) \right)$, we can conclude with the above Lemma. ■

Proposition 3.10. Let $n = r + s$ for $r, s \in \mathbb{N}$, then $\mathrm{CT}_{r, s}^{\mathrm{Coh}}$ is conservative on $\mathcal{P}_n^{\mathrm{Coh}}$.

Proof. We want to prove for any $\mathbf{P} \in \mathcal{P}_{\alpha}^{\mathrm{Coh}}$ with $\mathrm{rk}(\alpha) = n$, we can find $\mathbf{Q} \in \mathcal{Q}_{r, s}^{\mathrm{Coh}}$ such that

$$\left\langle \mathbf{Q}, (\mathrm{Ps}-\mathrm{Id}_!)^{-1} \left(\mathrm{CT}_{r, s}^{\mathrm{Coh}}(\mathbf{P}) \right) \right\rangle \neq 0$$

By Proposition 2.13, this is equivalent to

$$\left\langle \mathbf{Q}, \left(\mathrm{CT}_{r, s}^{\mathrm{Coh}} \right)_{\mathrm{co}, *} \left((\mathrm{Ps}-\mathrm{Id}_!)^{-1}(\mathbf{P}) \right) \right\rangle = \left\langle \mathrm{Eis}_{r, s}^{\mathrm{Coh}}(\mathbf{Q}), (\mathrm{Ps}-\mathrm{Id}_!)^{-1}(\mathbf{P}) \right\rangle \neq 0$$

Since the pairing is defined using Hom , we know for any $\mathbf{P}'$ over $\mathbf{Coh}_r$,

$$\left\langle \mathbf{P}', (\mathrm{Ps}-\mathrm{Id}_!)^{-1}(\mathbf{P}') \right\rangle = \mathrm{Hom} \left((\mathbf{P}')^{\vee}, (\mathrm{Ps}-\mathrm{Id}_!)^{-1}(\mathbf{P}') \right) \neq 0$$

By definition, $\mathbf{P} \subseteq \mathrm{Eis}_{\underline{\alpha}}^{\mathrm{Coh}} (\mathbf{1}_{\underline{\alpha}})$ for principal $\underline{\alpha}$. There exists a subpartition $\underline{\alpha} = \underline{\beta} \cup \underline{\gamma}$, such that the rank of $\beta = \sum \underline{\beta}$ and $\gamma = \sum \underline{\gamma}$ is r and s respectively. We decompose $\mathrm{Eis}_{\underline{\alpha}}^{\mathrm{Coh}}$ into composition $\mathrm{Eis}_{\underline{\beta}, \underline{\gamma}}^{\mathrm{Coh}} \circ \mathrm{Eis}_{\underline{\alpha}}^{\mathrm{Coh}, \underline{\beta}, \underline{\gamma}}$, then the complex $\mathrm{Eis}_{\underline{\alpha}}^{\mathrm{Coh}, \underline{\beta}, \underline{\gamma}} (\mathbf{1}_{\underline{\alpha}}) \in D(\mathbf{Coh}_{r, s})$ is semisimple and $\mathbf{P} \subseteq \mathrm{Eis}_{\underline{\beta}, \underline{\gamma}}^{\mathrm{Coh}} \circ \mathrm{Eis}_{\underline{\alpha}}^{\mathrm{Coh}, \underline{\beta}, \underline{\gamma}} (\mathbf{1}_{\underline{\alpha}})$, so we can take $\mathbf{Q}$ to be the subcomplex in $\mathrm{Eis}_{\underline{\alpha}}^{\mathrm{Coh}, \underline{\beta}, \underline{\gamma}} (\mathbf{1}_{\underline{\alpha}})$ such that $\mathrm{Eis}_{\underline{\beta}, \underline{\gamma}}^{\mathrm{Coh}}(\mathbf{Q}) \supseteq \mathbf{P}$. ■

3.2 Purity and trace map

We assume that $X/\overline{\mathbb{F}}_q$ is defined over $\mathbb{F}_q$, meaning $X = X_0 \times_{\mathbb{F}_q} \overline{\mathbb{F}}_q$ for some smooth projective curve X_0 over $\mathbb{F}_q$. Over curve X_0 , there is a representable Quot scheme ${}^0\text{Quot}_\alpha^\mathcal{L}$ and its open ${}^0\mathcal{Q}_\alpha^\mathcal{L}$ whose underlying functors are similarly defined as $\text{Quot}_\alpha^\mathcal{L}$ and $\mathcal{Q}_\alpha^\mathcal{L}$, so that we can see the quasi-compact open covers $\mathbf{Coh}_\alpha^\mathcal{L}$ are defined over $\mathbb{F}_q$, namely we have $\mathbf{Coh}_\alpha^{>\mathcal{L}} \cong {}^0\mathbf{Coh}_\alpha^{>\mathcal{L}} \times_{\mathbb{F}_q} \overline{\mathbb{F}}_q$ and $\mathcal{Q}_\alpha^\mathcal{L} \cong {}^0\mathcal{Q}_\alpha^\mathcal{L} \times_{\mathbb{F}_q} \overline{\mathbb{F}}_q$. And thus there is a geometric Frobenius action defined over $\mathbf{Coh}_\alpha$. Recall any object $\mathbf{P}$ in $\text{Shv}(\mathbf{Coh}_\alpha)$ is called a Weil sheaf if there is an isomorphism $\mathbf{P} \xrightarrow{\cong} (\text{Frob})^*(\mathbf{P})$.

Proposition 3.11. [Sch06] *When genus of X is less than 2, any $\mathbf{P} \in \mathcal{P}_\alpha^{\text{Coh}}$ is a Weil sheaf that pointwise pure of weight zero.*

Proof. We refer to the proof in [Sch06]. It is a consequence of three facts. First, we have $\mathbf{1}_\alpha$ is a lisse sheaf pointwise pure of weight zero. Due the semisimplicity of $\text{Eis}_\alpha^{\text{Coh}}(\mathbf{1}_\alpha)$, it is written as $\oplus \mathbb{V}_\mathbf{P} \oplus \mathbf{P}$, where for each $\mathbf{P}$, the complex $\mathbb{V}_\mathbf{P} = \oplus_j \text{Hom}\left(\mathbf{P}, {}^p H^j\left(\text{Eis}_\alpha^{\text{Coh}}(\mathbf{1}_\alpha)\right)\right)[j]$ is of weight zero. Moreover, we have pointwise $H^k\left(\text{Eis}_\alpha^{\text{Coh}}(\mathbf{1}_\alpha)\right)|_{\mathcal{F}} = \bigoplus_{\mathbf{P}} \bigoplus_{i+j=k} H^i(\mathbb{V}_\mathbf{P}) \otimes H^j(\mathbf{P})|_{\mathcal{F}}$, thus we can deduce $\mathbf{P}$ is pointwise pure of weight zero from the assertion that $\text{Eis}_\alpha^{\text{Coh}}(\mathbf{1}_\alpha)$ is pointwise pure of weight zero. It remains to show this assertion for any principal series tuple $\underline{\alpha} = (\alpha_1, \dots, \alpha_s)$.

For any $\mathcal{F} \in \mathbf{Coh}_\alpha$, there is a line bundle $\mathcal{L}$ such that $\mathcal{F}$ locates in $[\mathcal{Q}_\alpha^\mathcal{L}/G_\alpha^\mathcal{L}]$. We use $\mathbb{O}_{\mathcal{F}}^\mathcal{L}$ to denote the $G_\alpha^\mathcal{L}$ -orbit of $\mathcal{F}$ in $[\mathcal{Q}_\alpha^\mathcal{L}/G_\alpha^\mathcal{L}]$. There is a bijection between isomorphism classes in $\mathbf{Coh}_\alpha(\mathbf{k})$ that strongly generated by $\mathcal{L}$ and $G_\alpha^\mathcal{L}$ -orbits in $\mathcal{Q}_\alpha^\mathcal{L}$. We can find a point $z \in \mathbb{O}_{\mathcal{F}}^\mathcal{L}$ and an integer e such that $(\text{Frob}^*)^e(z) = z$. Thus we reduce to show that for any z' in the orbit $\mathbb{O}_{\mathcal{F}}^\mathcal{L}$, the eigenvalue of $(\text{Frob}^*)^e$ on $H^i(\text{Eis}_{\underline{\alpha}})|_{z'}$ has norm $q^{\frac{ei}{2}}$. We consider the space $E_{\underline{\alpha}}^\mathcal{L}$ over $\mathcal{Q}_\alpha^\mathcal{L}$, given by

$$\left\{ (\phi, V_\bullet = V_s \subseteq V_{s-1} \subseteq \dots \subseteq V_1) \left| \phi : \mathbf{k}^{(c(\mathcal{L}), \alpha)} \twoheadrightarrow \mathcal{F}; \begin{array}{l} \dim(V_i/V_{i+1}) = \langle \mathcal{L}, \alpha_i \rangle \\ \phi(V_i \otimes \mathcal{L}) / \phi(V_{i+1} \otimes \mathcal{L}) \in \mathbf{Coh}_{\alpha_i}(\mathbf{k}) \end{array} \right. \right\}$$

We consider first the points in very unstable locus of $\mathbf{Coh}_\alpha$. Let $\mathcal{F} \in \mathbf{Coh}_\alpha^{\text{vu}}(\mathbf{k})$ be such a point, . . . ■

Proposition 3.12. *The trace of Frobenius on $\mathcal{Q}^{\text{Coh}}$ commutes with Eis^{Coh} and CT^{Coh} .*

4 Lusztig sheaves and Hall algebras

4.1 The Hall algebra of curve and trace of Laumon-Lusztig category

When we work on X is a smooth projective curve over $\overline{\mathbb{F}}_q$, we assume in this section that X is defined over $\mathbb{F}_q$, namely there exists X_0 over $\mathbb{F}_q$ such that $X = X_0 \times_{\mathbb{F}_q} \overline{\mathbb{F}}_q$. We briefly recall the definition of Hall algebra of X_0 and its spherical part. Denoted by $\text{Coh}(X_0)$ the category of coherent sheaves on X_0 , it has locally finite isomorphism classes and $\text{Tor}(X_0) \subseteq \text{Coh}(X_0)$ subcategory of torsion sheaves on X_0 .

We fix an identification $\overline{\mathbb{Q}}_\ell \cong \mathbb{C}$ and a square root v of q^{-1} , the **Hall algebra** $\mathbf{H}_{X_0}$ of X_0 has underlying vector space spanned by isomorphism classes $\bigoplus_{[\mathcal{F}] \in \text{Coh}(X_0)} \mathbb{C}[\mathcal{F}]$, it carries a bialgebra structure with multiplication given by $[\mathcal{F}] * [\mathcal{G}] = \sum_{\mathcal{M} \in \text{Ext}^1(\mathcal{F}, \mathcal{G})} v^{-\langle \mathcal{F}, \mathcal{G} \rangle} [\mathcal{M}]$ and comultiplication in a reverse way, see [Sch09] for more details. Thus, the multiplication and comultiplication in Hall algebra can be viewed as decategorification of Eis^{Coh} and CT^{Coh} . As a vector space, $\mathbf{H}_{X_0}$ can also be identified with

$$\mathbf{H}_{X_0} = \left(\bigoplus_{\alpha \in \mathbb{Z}^+} \text{Funct}_0({}^0\mathbf{Coh}_\alpha(\mathbb{F}_q), \mathbb{C}) \right) \otimes \mathbb{C}[\kappa^{\pm 1}]$$

where κ satisfies $\kappa f \kappa^{-1} = v^{2r(g_X-1)} f$ for any $f \in \text{Funct}_c({}^0\mathbf{Coh}_r(\mathbf{k}), \mathbb{C})$. For $f, g \in \mathbf{H}_{X_0}$, we have multiplication and coproduct in $\mathbf{H}_{X_0}$ given concretely by

$$\begin{aligned} f * g(\mathcal{M}) &= \sum_{\mathcal{N} \subseteq \mathcal{M}} v^{-\langle \mathcal{M}/\mathcal{N}, \mathcal{N} \rangle} f(\mathcal{M}/\mathcal{N}) g(\mathcal{N}) \\ \Delta(\kappa) &= \kappa \otimes \kappa; \quad \Delta_{r,s}(f) = \sum_{\mathcal{J}, \mathcal{K}} \frac{v^{\langle \mathcal{J}, \mathcal{K} \rangle}}{|\text{Ext}^1(\mathcal{M}, \mathcal{N})|} \sum_{\xi \in \text{Ext}^1(\mathcal{J}, \mathcal{K})} f(\mathcal{M}_\xi) \mathbf{1}_{[\mathcal{J}]} \kappa^s \otimes \mathbf{1}_{[\mathcal{K}]} \end{aligned}$$

where $\mathbf{1}_{[\mathcal{J}]}$ is the characteristic function of the class $[\mathcal{J}]$ and $\mathcal{M}_\xi$ is the class corresponds to the extension $\xi \in \text{Ext}^1(\mathcal{J}, \mathcal{K})$ and the sum for $\Delta_{r,s}$ is taken over $\mathcal{J}, \mathcal{K}$ of rank r and s respectively.

Definition 4.1. The **spherical Hall algebra** $\mathbf{H}_{X_0}^{\text{sph}}$ of X_0 is defined to be the subalgebra of $\mathbf{H}_{X_0}$, which is topological completion of the space generated by $\mathbf{1}_\alpha$ and $\kappa^{\pm 1}$ under the product in $\mathbf{H}_{X_0}$, where $\text{rk}(\alpha) \leq 1$ is principal and $\mathbf{1}_\alpha$ denotes the characteristic function on ${}^0\mathbf{Coh}_\alpha(\mathbb{F}_q)$.

For $\alpha \in \mathbb{Z}^+$, we denote by $\mathbf{H}_{X_0}[\alpha]$ the part supported on $e([\mathcal{F}]) = \alpha$, for a tuple $\underline{\alpha} = (\alpha_1, \dots, \alpha_s) \in (\mathbb{Z}^+)^s$, we use $\mathbf{H}_{X_0}[\underline{\alpha}]$ to denote for $\mathbf{H}_{X_0}[\alpha_1] \widehat{\boxtimes} \dots \widehat{\boxtimes} \mathbf{H}_{X_0}[\alpha_s]$. The same notions go for $\mathbf{H}_{X_0}^{\text{sph}}$.

Proposition 4.2. There is a bialgebra structure on $\mathbf{H}_{X_0}^{\text{sph}}$ with a Hopf pairing given by

$$\langle \mathbf{1}_{\mathcal{M}} \kappa^r, \mathbf{1}_{\mathcal{N}} \kappa^s \rangle := \frac{\delta_{\mathcal{M}, \mathcal{N}}}{|\text{Aut}(\mathcal{M})|} v^{2(g_X-1)rs}$$

Inside of $\mathbf{H}_{X_0}^{\text{sph}}$, we have two subalgebras $\mathbf{H}_{X_0}^{\text{sph}, \text{Bun}}$, $\mathbf{H}_{X_0}^{\text{sph}, \text{Tor}}$ which are generated by $\{\mathbf{1}_{(1,d)}\}_{d \in \mathbb{Z}}$ and $\{\mathbf{1}_{\delta l}\}_{l \geq 1}$

respectively.

On the other hand, we consider the algebra coming from Laumon-Lusztig category:

$$\hat{\mathbf{u}}_X^+ = \text{Tr} \left(\text{Frob}, \bigoplus_{\alpha \in \mathbb{Z}^+} Q_\alpha^{\text{Coh}} \right)$$

as a vector space, it contains trace of Frobenius over Q^{Coh} . By Proposition 3.12, the functors Eis^{Coh} and CT^{Coh} endows $\hat{\mathbf{u}}_X^+$ with structures of multiplication and comultiplication respectively. We use the same notation Eis^{Coh} and CT^{Coh} for operations in $\hat{\mathbf{u}}_X^+$. Moreover, the pairing on $\text{Shv}(\mathbf{Coh}_\alpha)$ (see section 1.2) restricts to Q_α^{Coh} and endows $\hat{\mathbf{u}}_X^+$ a non-degenerate pairing that compatible with Eis^{Coh} and CT^{Coh} . For $\mathbf{P} \in Q^{\text{Coh}}$, we denote by $f_{\mathbf{P}} \in \hat{\mathbf{u}}_X^+ \subseteq \text{Funct}(\mathbf{Coh}(\mathbb{F}_q), \overline{\mathbb{Q}}_\ell)$. In fact, we have the local term map $\text{LT} : \text{Tr}(\text{Frob}, \text{Shv}(\mathbf{Coh}_\alpha)) \longrightarrow \text{Funct}(\mathbf{Coh}_\alpha(\mathbb{F}_q), \overline{\mathbb{Q}}_\ell)$ and we have the following Lemma:

Lemma 4.3. (i) The restriction of local term map to Q^{Coh} gives an inclusion $\hat{\mathbf{u}}_X^+ \subseteq \mathbf{H}_{X_0}$.
(ii) Let $\langle -, - \rangle : \hat{\mathbf{u}}_X^+ \otimes \hat{\mathbf{u}}_X^+ \longrightarrow \mathbb{C}$ be the natural pairing in function spaces, then it satisfies

$$\langle f_{\mathbf{P}_1}, f_{\mathbf{P}_2} \rangle := \langle \mathbf{P}_1, (\text{Ps-Id}_!)^{-1}(\mathbf{P}_2) \rangle$$

and we have $\langle \text{Eis}_{\underline{\alpha}}^{\text{Coh}}(f), g \rangle = \langle f, \text{CT}_{\underline{\alpha}}^{\text{Coh}}(g) \rangle$ for $f \in \hat{\mathbf{u}}_X^+[\underline{\alpha}]$, $g \in \hat{\mathbf{u}}_X^+[\alpha]$.

We have the following main Theorem of this section, the proof will be given in 4.4.

Theorem 4.4. The above inclusion $\hat{\mathbf{u}}_X^+ \subseteq \mathbf{H}_{X_0}$ identifies $\hat{\mathbf{u}}_X^+$ with the subalgebra $\mathbf{H}_{X_0}^{\text{sph}}$.

4.2 The case of torsion sheaves and Hecke action

In this section, we give a more explicit description of the rank 0 part of $\hat{\mathbf{u}}_X^+$. In fact, we have subcategory $Q_0^{\text{Coh}} \subseteq Q^{\text{Coh}}$ spanned by objects in $\mathcal{P}^{\text{Tor}}$, which is closed under Eis^{Coh} , CT^{Coh} . Taking trace of Frobenius of objects in Q_0^{Coh} , we get a sub(co)-algebra $\hat{\mathbf{u}}_X^+[0]$ of $\hat{\mathbf{u}}_X^+$. The Springer theory helps us to determine explicitly the category Q_0^{Coh} as well as $\hat{\mathbf{u}}_X^+[0]$.

For fixed torsion degree $d \geq 1$, we have the following diagram of Abel-Jacobi maps and induction:

$$\begin{array}{ccccc} \mathbf{Coh}_{\tau 1} \times \dots \times \mathbf{Coh}_{\tau 1} & \xleftarrow{q_{\tau 1}^{\text{Coh}}} & \widetilde{\mathbf{Coh}}_{\tau 1} & \xrightarrow{q_{\tau 1}^{\text{Coh}}} & \mathbf{Coh}_{\tau d} \\ & \searrow \text{supp}_{\times d} & \begin{array}{c} \widetilde{\text{AJ}}^d \updownarrow \text{supp}_d \\ \text{AJ}^d \updownarrow \text{supp}_d \end{array} & & \begin{array}{c} \text{AJ}^d \updownarrow \text{supp}_d \\ \text{AJ}^d \updownarrow \text{supp}_d \end{array} \\ & & X^d & \xrightarrow{\text{Sym}} & X^{(d)} \end{array}$$

where $\tau 1$ is d -copies of $\tau 1 = (0, 1)$. Inside of X^d (resp. $X^{(d)}$), consider the open locus $X^d \setminus \Delta$ (resp. $X^{(d)} \setminus \Delta$) of points with multiplicities 1. Then the Abel Jacobi map gives rises to an open inclusion $(X^{(d)} \setminus \Delta) / (\mathbb{G}_m)^d =: \mathbf{Coh}_{\tau d}^{\text{rss}} \subseteq \mathbf{Coh}_{\tau d}$, where the $(\mathbb{G}_m)^d$ -action is trivial and the upper script rss refers

to regular semisimple. In fact, recall when we take X to be $\mathbb{A}^1$, $\mathbf{Coh}_{\tau d} \cong [\mathfrak{gl}_d/\mathrm{GL}_d]$, the map $q_{\tau 1}^{\mathrm{Coh}}$ is the Grothendieck-Springer resolution $[\widetilde{\mathfrak{gl}}_d/\mathrm{GL}_d] \rightarrow [\mathfrak{gl}_d/\mathrm{GL}_d]$, the preimage of supp_d (resp. $\widetilde{\mathrm{supp}}_d$) corresponds to the regular semisimple part $[\mathfrak{gl}_d^{\mathrm{rss}}/\mathrm{GL}_d]$ (resp. $[\widetilde{\mathfrak{gl}}_d^{\mathrm{rss}}/\mathrm{GL}_d]$). The restriction of $q_{\tau 1}^{\mathrm{Coh}}$ to $[\widetilde{\mathfrak{gl}}_d^{\mathrm{rss}}/\mathrm{GL}_d]$ is an étale $\mathfrak{S}_d$ -cover. In general, for X a smooth projective curve, the map $X^d \setminus \Delta \rightarrow X^{(d)} \setminus \Delta$ is also a $\mathfrak{S}_d$ -cover. A representation σ of $\mathfrak{S}_d$ thus gives rises to a local system $\mathbf{L}_\sigma$ on $\mathbf{Coh}_{\tau d}^{\mathrm{rss}}$.

Proposition 4.5. *For X smooth projective curve, the simple objects in $\mathcal{P}_{\tau d}^{\mathrm{Tor}}$ are in bijection with*

$$\{\mathbf{IC}(\mathbf{Coh}_{\tau d}^{\mathrm{rss}}, \mathbf{L}_\sigma) \mid \sigma : \text{irreducible representation of } \mathfrak{S}_d\}.$$

In particular, we know irreducible representations of $\mathfrak{S}_d$ are classified by Young diagrams of size d . We use $\mathrm{Part}(d)$ to denote the set of Young diagrams of size d and σ_λ for the irreducible representation that is associated with $\lambda \in \mathrm{Part}(d)$. We abbreviate $\mathbf{L}_\lambda$ the object $\mathbf{IC}(\mathbf{Coh}_{\tau d}^{\mathrm{rss}}, \mathbf{L}_{\sigma_\lambda})$ in $\mathcal{P}_{\tau d}^{\mathrm{Tor}}$ that corresponds to irreducible representation σ_λ . Now for $d_1, d_2 \geq 1$ and $d = d_1 + d_2$, we have the following relation

$$\mathrm{Eis}_{\tau d_1, \tau d_2}^{\mathrm{Coh}}(\mathbf{L}_{\lambda_1} \boxtimes \mathbf{L}_{\lambda_2}) \cong \bigoplus_{\lambda \in \mathrm{Part}(d)} \mathbb{V}_\lambda \otimes \mathbf{L}_\lambda[\mathrm{shift}]$$

where $\mathbb{V}_\lambda$ is vector space $\mathbb{C}^{c_{\lambda_1, \lambda_2}^\lambda}$ of dimension $c_{\lambda_1, \lambda_2}^\lambda$, the multiplicity of σ_λ that appears in the induced representation $\mathrm{Ind}_{\mathfrak{S}_{d_1} \times \mathfrak{S}_{d_2}}^{\mathfrak{S}_d}(\sigma_{\lambda_1} \boxtimes \sigma_{\lambda_2})$.

On the other hand, we consider $\mathbf{H}_{X_0}^{\mathrm{Tor}} := \mathbf{H}_{X_0}[0]$, which is by definition the space of functions of compact support over ${}^0\mathbf{Coh}^{\mathrm{Tor}}(\mathbb{F}_q)$ or equivalently function of objects in $\mathrm{Tor}(X_0)$. Since $\mathrm{Tor}(X_0)$ is a Serre subcategory of $\mathrm{Coh}(X_0)$, $\mathbf{H}_{X_0}^{\mathrm{Tor}}$ is a sub-bialgebra of the Hall algebra $\mathbf{H}_{X_0}$. It is not hard to see that the bialgebra $\mathbf{H}_{X_0}^{\mathrm{Tor}}$ is isomorphic to a restricted tensor product $\left(\bigotimes'_{x \in X} \mathbf{H}_x^{\mathrm{Tor}}\right) \otimes \mathbb{C}[\kappa^{\pm 1}]$, where $\mathbf{H}_x^{\mathrm{Tor}}$ is the function on torsion sheaves that supported on x . In fact, $\mathbf{H}_x^{\mathrm{Tor}}$ is known to be isomorphic to the classical Hall algebra. Correspondingly, we consider the spherical part of $\mathbf{H}_{X_0}^{\mathrm{Tor}}$ that generated by $\{\mathbf{1}_{\tau d}\}_{d \geq 1}$, which we denote by $\mathbf{H}_{X_0}^{\mathrm{sph, Tor}} \subseteq \mathbf{H}_{X_0}^{\mathrm{sph}}$. Moreover, it is known that

Proposition 4.6. *The collection $\bigsqcup_{d \geq 1} \{f_{\mathbf{L}_\lambda} \mid \lambda \in \mathrm{Part}(d)\}$ gives rises to a basis of $\mathbf{H}_{X_0}^{\mathrm{sph, Tor}}$. In particular the algebra $\hat{\mathbf{U}}_X^+[0]$ that spanned by $f_{\mathbf{L}_\lambda}$ is isomorphic to $\mathbf{H}_{X_0}^{\mathrm{sph, Tor}}$.*

There is an action of $\mathbf{H}_{X_0}^{\mathrm{Tor}}$ on $\mathbf{H}_{X_0}$ (referred as Hecke action in [Kap96]), which is given by

$$\mathbf{H}_{X_0}^{\mathrm{Tor}} \times \mathbf{H}_{X_0} \longrightarrow \mathbf{H}_{X_0}; (\mathbf{1}_\mathcal{T}, f) \mapsto \left(T_\mathcal{T} \cdot f : \mathcal{E} \mapsto \sum_{\mathcal{E}' \subseteq \mathcal{E}, \mathcal{E}/\mathcal{E}' \cong \mathcal{T}} f(\mathcal{E}') \right)$$

It can be regarded as decategorification of geometric Hecke action, note that for fixed $\alpha \in \mathbf{Z}^{++}$ and torsion degree d , the geometric Hecke action is given by pull-push along the Hecke stack $\mathbf{Hk}_\alpha^d$ that

classifies $\langle 0 \rightarrow \mathcal{E}' \rightarrow \mathcal{E} \rightarrow \mathcal{T} \rightarrow 0 \mid \mathcal{E}' \in \mathbf{Bun}_\alpha, \mathcal{E} \in \mathbf{Bun}_{\alpha+\tau d} \rangle$. We have following diagram:

$$\begin{array}{ccccc}
\mathbf{Coh}_{\tau d} \times \mathbf{Bun}_\alpha & \xleftarrow{h^\leftarrow} & \mathbf{Hk}_\alpha^d & \xrightarrow{h^\rightarrow} & \mathbf{Bun}_{\alpha+\tau d} \\
\downarrow & & \downarrow & & \downarrow \\
\mathbf{Coh}_{\tau d} \times \mathbf{Coh}_\alpha & \xleftarrow{q_{\alpha,\tau d}^{\text{Coh}}} & \widetilde{\mathbf{Coh}}_{\alpha,\tau d} & \xrightarrow{p_{\alpha,\tau d}^{\text{Coh}}} & \mathbf{Coh}_{\alpha+\tau d}
\end{array}$$

On the spherical part, the above Hecke action on $\mathbf{Bun}$ restricts to the action of $\mathbf{H}_{X_0}^{\text{sph},\text{Tor}}$ on another subalgebra $\mathbf{H}_{X_0}^{\text{sph},\text{Bun}}$ of $\mathbf{H}_{X_0}^{\text{sph}}$. In particular, $\mathbf{H}_{X_0}^{\text{sph},\text{Tor}}$ can be regarded as *global spherical Hecke algebra* and the algebra $\mathbf{H}_{X_0}^{\text{sph}}$ is generated by $\mathbf{H}_{X_0}^{\text{sph},\text{Tor}}$ and $\mathbf{H}_{X_0}^{\text{sph},\text{Bun}}$ modulo Rankin-Selberg relation. The basis $\{f_{\mathbf{L}_\lambda}\}$ we obtained in Proposition 4.5 is thus an analog of Kazhdan-Lusztig basis for curve.

4.3 Canonical bases associated with curves

In [Sch06] and [Sch12], Schiffmann considered the canonical bases for quantum algebra that can be reconstructed from curves. For $X_0 = \mathbb{P}^1$, the spherical Hall algebra $\mathbf{H}_{X_0}^{\text{sph}}$ gives $\mathbf{U}_v^+(\hat{\mathfrak{sl}}_2)$, the positive half of quantum affine $\mathfrak{sl}_2$. When X_0 is an elliptic curve, the spherical Hall algebra $\mathbf{H}_{X_0}^{\text{sph}}$ is a deformation of the ring of diagonal symmetric polynomials $\mathbf{M}^+ = \mathbb{C}[x_i^{\pm 1}, \dots, y_i]^{\mathfrak{S}_\infty}$, where the deformation is given by two parameters $\sigma, \bar{\sigma}$ that correspond to the Weil numbers of X_0 . Moreover, there is *universal elliptic Hall algebra* $\mathcal{E}_{\mathbf{R}}^+$ over $\mathbf{R} = \mathbb{C}[\sigma^{\pm \frac{1}{2}}, \sigma'^{\pm \frac{1}{2}}]$ that realizes $\mathbf{H}_{X_0}^{\text{sph}}$ of each elliptic curve X_0 as a specialization of σ, σ' to Weil numbers. More interestingly, there is projection $\mathcal{E}_{\mathbf{R}}^+ \rightarrow \mathbf{SH}_n^+$ from universal elliptic Hall algebra to spherical double affine Hecke algebra (DAHA) associated with any $n \geq 1$.

The geometric way to construct canonical bases proposed by Schiffmann is through Eisenstein sheaves. More concretely, We can consider $K_0(\mathcal{Q}^{\text{Coh}}) := \bigoplus_{\alpha \in \mathbb{Z}^+} K_0(\mathcal{Q}_\alpha^{\text{Coh}})$, the Grothendieck ring of (Karoubi complete) Laumon-Lusztig category. Since $\mathcal{Q}^{\text{Coh}}$ is stable under Eis^{Coh} and CT^{Coh} , we can endow $K_0(\mathcal{Q}^{\text{Coh}})$ with multiplication and comultiplication structures. In fact, $K_0(\mathcal{Q}^{\text{Coh}})$ is a bialgebra and we use the notation $(K_0(\mathcal{Q}^{\text{Coh}}), \cdot, \Delta)$ for $\mathcal{Q} \in \mathcal{Q}^{\text{Coh}}$ for those two structures on it. We denote by $\mathbf{b}_{\mathcal{Q}}$ the corresponding element in $K_0(\mathcal{Q}^{\text{Coh}})$.

Lemma 4.7. *Similar to Lemma 4.3(ii), there is a Hopf pairing on $K_0(\mathcal{Q}^{\text{Coh}})$ given by*

$$\langle \mathbf{b}_{\mathbf{P}_1}, \mathbf{b}_{\mathbf{P}_2} \rangle := \langle \mathbf{P}_1, (\text{Ps-Id}_!)^{-1}(\mathbf{P}_2) \rangle$$

and we have $\langle \mathbf{b}_1 \cdot \mathbf{b}_2, \mathbf{b}_3 \rangle = \langle \mathbf{b}_1 \otimes \mathbf{b}_2, \Delta(\mathbf{b}_3) \rangle$ for $\mathbf{b}_i \in K_0(\mathcal{Q}_{\alpha_i}^{\text{Coh}})$, $i = 1, 2$ and $\mathbf{b}_3 \in K_0(\mathcal{Q}_{\alpha_1+\alpha_2}^{\text{Coh}})$.

Definition 4.8. We use the notation $\mathbf{B}_X := \{\mathbf{b}_{\mathbf{P}}\}_{\mathbf{P} \in \mathcal{P}^{\text{Coh}}}$ for the subset in $K_0(\mathcal{Q}^{\text{Coh}})$ that labelled by simple Eisenstein sheaves in $\mathcal{P}^{\text{Coh}}$, we refer $\mathbf{B}_X$ as the set of **canonical bases** associated with X .

For any degree $e \geq 1$, the trace of Frobenius (of degree e) induces a map $\mathbf{B}_X \mapsto \{f_{\mathbf{P}}^e\}_{\mathbf{P} \in \mathcal{P}^{\text{Coh}}}$, where $f_{\mathbf{P}}^e$ is a function in $\mathbf{H}_{X/\mathbb{F}_q^e}$: the Hall algebra associated with $X/\mathbb{F}_q^e$.

We have the following results of Schiffmann [Sch06] [Sch12] concerning the case of genus 0 and 1.

Theorem 4.9. (i) When $g_X = 1$, namely X_0 is $\mathbb{P}^1$, the (trace of) set $\mathbf{B}_X$ gives a canonical basis for $\mathbf{U}_v^+(\hat{\mathfrak{sl}}_2)$.

(ii) When $g_X = 1$, the set $\mathbf{B}_X$ gives an $\mathbf{R}$ -basis of universal Elliptic Hall algebra $\mathcal{E}_{\mathbf{R}}^+$. Moreover, it shares the following properties with canonical bases of quantum algebra:

- (Convolution) The Verdier duality induces an isomorphism $\overline{\mathbf{b}_{\mathbf{P}}} := \mathbf{b}_{\mathbb{D}(\mathbf{P})} \cong \mathbf{b}_{\mathbf{P}}$, for any $\mathbf{P} \in \mathcal{P}^{\text{Coh}}$.
- (Positivity) For $\mathbf{P}, \mathbf{P}' \in \mathcal{P}^{\text{Coh}}$, we have $\mathbf{b}_{\mathbf{P}} \cdot \mathbf{b}_{\mathbf{P}'} \in \prod_{\mathbf{P}''} \mathbb{N}[v^{\pm 1}] \mathbf{b}_{\mathbf{P}''}$.

As a corollary of Theorem 4.4, we can generalize the result of Schiffmann to curve of higher genus.

Corollary 4.10. Let X_0 be of any genus, then for each degree $e \geq 1$, the set $\mathbf{B}_X$ gives rises to a basis of spherical Hall algebra $\mathbf{H}_{X/\mathbb{F}_q^e}$ by taking trace of Frobenius of degree e . Moreover, $\mathbf{B}_X$ satisfies the same positivity as in Theorem 4.9 and the Verdier duality induces an auto-equivalence on $\mathbf{B}_X$.

Also, we can consider the space of principal series automorphic forms $\mathbf{H}_{X_0}^{\text{sph}, \text{Bun}} \subseteq \mathbf{H}_{X_0}^{\text{sph}}$. It is shown in [NSS25] that when Weil numbers of X_0 are different, $\mathbf{H}_{X_0}^{\text{sph}, \text{Bun}}$ is isomorphic to the (small) shuffle algebra associated with quiver of g_X -loops, see also [Neg23]. Combine with Theorem 3.5, we obtain the canonical bases for quantum loop algebra $\mathbf{H}_{X_0}^{\text{sph}, \text{Bun}}$ via compactified Eisenstein series.

Corollary 4.11. The set $\{\mathbf{b}_{\mathbf{P}}\}_{\mathbf{P} \in \bigsqcup_r \mathcal{P}_r^{\text{Bun}}}$ forms a canonical basis of $\mathbf{H}_{X_0}^{\text{sph}, \text{Bun}}$.

Proof. According to Proposition 4.5, the objects in $\mathcal{P}^{\text{Tor}}$ are of the form $\mathbf{IC}(\mathbf{Coh}_{\tau d}^{\text{rss}}, \mathbf{L}_{\sigma})$, hence self-dual. Then we recall for $\mathbf{P} \in \mathcal{P}^{\text{Coh}}$, we can associate it with $(\mathbf{P}^{\text{Bun}}, \mathbf{T}) \in \mathcal{P}^{\text{Bun}} \times \mathcal{P}^{\text{Tor}}$ and in the proof of Theorem 3.5, we give a construction that $\mathbf{P} \cong (\mathbf{P}^{\text{Bun}} \boxtimes \mathbf{T})^{\text{Coh}}$. We can calculate the Verdier dual:

$$\begin{aligned} \mathbb{D} \left((\mathbf{P}^{\text{Bun}} \boxtimes \mathbf{T})^{\text{Coh}} \right) &= \mathbb{D} \left(\text{Eis}^{\text{Coh}} ((j_{\alpha} \times \text{Id})_{!*} (\mathbf{P}^{\text{Bun}} \boxtimes \mathbf{T})) \right) \\ &= \text{Eis}^{\text{Coh}} (\mathbb{D} ((j_{\alpha} \times \text{Id})_{!*} (\mathbf{P}^{\text{Bun}} \boxtimes \mathbf{T}))) \\ &= \text{Eis}^{\text{Coh}} ((j_{\alpha} \times \text{Id})_{!*} (\mathbb{D} (\mathbf{P}^{\text{Bun}}) \boxtimes \mathbb{D} (\mathbf{T}))) = (\mathbf{P}^{\text{Bun}} \boxtimes \mathbf{T})^{\text{Coh}} \end{aligned}$$

■

4.4 Proof of Theorem 4.4

By definition and Proposition 4.6, we have isomorphisms $\hat{\mathbf{u}}_X^+[i] := \text{Tr} \left(\text{Frob}, \bigoplus_{\text{rk}(\alpha)=i} \mathcal{Q}_\alpha^{\text{Coh}} \right) \cong \mathbf{H}_{X_0}^{\text{sph}}[i]$ for $i = 0, 1$. We would use Proposition 3.12 to translate sheaves to functions. Assume first that $\mathbf{P} \in \mathcal{P}_\alpha^{\text{Coh}}$ is a simple object, we want to show when $\mathbf{P}$ is supported only on very unstable part, the corresponding function $f_{\mathbf{P}} \in \mathbf{H}_{X_0}^{\text{sph}}[\alpha]$. The following Lemma is crucial, which gives a bound to the support for Eisenstein sheaves.

Lemma 4.12. *Let $\underline{\beta} = (\beta_1, \dots, \beta_m)$ be a tuple of HN-type, then for any $\mathcal{F} \in \mathbf{Coh}_{\underline{\beta}}$, we have*

$$\mathbf{p}_{\underline{\beta}}^{\text{Coh}} \left(\mathbf{q}_{\underline{\beta}}^{\text{Coh}} \right)^{-1} (\mathcal{F}) \subseteq \bigcup_{\underline{\gamma} \leq \underline{\beta}} \text{HN}^{-1}(\underline{\gamma})$$

as a consequence, the support of Eisenstein series $\text{supp} \left(\text{Eis}_{\underline{\beta}}^{\text{Coh}}(\mathbf{Q}) \right) \subseteq \bigcup_{\underline{\gamma} \leq \underline{\beta}} \text{HN}^{-1}(\underline{\gamma})$ for $\mathbf{Q} \in \mathcal{Q}_{\underline{\beta}}^{\text{Coh}}$.

By our assumption, $\text{supp}(\mathbf{P}) \cap \text{HN}^{-1}(\underline{\alpha}) \neq \emptyset$ for some very unstable $\underline{\alpha}$, we can take $\underline{\beta}$ to be the unique very unstable HN-type among those $\underline{\alpha}$ such that $\text{supp}(\mathbf{P}) \subseteq \overline{\text{HN}^{-1}(\underline{\beta})}$.

Lemma 4.13. *For the above very unstable $\underline{\beta}$ associated with $\mathbf{P}$, we have*

$$\mathbf{P}|_{\text{HN}^{-1}(\underline{\beta})} = \left(\text{Eis}_{\underline{\beta}}^{\text{Coh}} \circ \text{CT}_{\underline{\beta}}^{\text{Coh}}(\mathbf{P}) \right)|_{\text{HN}^{-1}(\underline{\beta})}$$

By induction hypothesis and the fact that $\text{CT}_{\underline{\beta}}^{\text{Coh}} : \mathcal{Q}_\alpha^{\text{Coh}} \rightarrow \mathcal{Q}_{\underline{\beta}}^{\text{Coh}}$ preserves Lusztig sheaf categories, we can deduce $f_{\text{CT}_{\underline{\beta}}^{\text{Coh}}(\mathbf{P})}$ lies in $\mathbf{H}^{\text{sph}}[\underline{\beta}]$, and thus $\text{Eis}_{\underline{\beta}}^{\text{Coh}} \left(f_{\text{CT}_{\underline{\beta}}^{\text{Coh}}(\mathbf{P})} \right) = f_{\text{Eis}_{\underline{\beta}}^{\text{Coh}} \circ \text{CT}_{\underline{\beta}}^{\text{Coh}}(\mathbf{P})} \in \mathbf{H}^{\text{sph}}$. On the other hand, abbreviate $\text{Eis}_{\underline{\beta}}^{\text{Coh}} \circ \text{CT}_{\underline{\beta}}^{\text{Coh}}(\mathbf{P})$ as $\mathbf{R}$, by assumption $\mathbf{P} \subseteq \mathbf{R}$ is a simple direct summand and there exists $\mathbf{P}'$ (not necessarily simple) such that $\mathbf{P} \oplus \mathbf{P}' = \mathbf{R}$ and that $\text{supp}(\mathbf{P}') \subseteq \bigcup_{\underline{\beta} < \underline{\alpha}} \text{HN}^{-1}(\underline{\alpha})$. Using the induction hypothesis, we have $f_{\mathbf{P}'} \in \mathbf{H}_{X_0}^{\text{sph}}[\alpha]$, so we can deduce $f_{\mathbf{P}} = f_{\mathbf{R}} - f_{\mathbf{P}'} \in \mathbf{H}_{X_0}^{\text{sph}}[\alpha]$. As a consequence, the space of traces of objects in $\mathcal{Q}_\alpha^{\text{Coh}}$ that having very unstable supports is included in $\mathbf{H}_{X_0}^{\text{sph}}[\alpha]$, denote it by $\mathbf{H}_{X_0}^{\text{sph}, \text{vus}}[\alpha]$.

Next we deal with the case where $\text{supp}(\mathbf{P}) \cap (\mathbf{Coh}_\alpha^{\text{nvu}} \cup \mathbf{Coh}_\alpha^{\text{as}}) \neq \emptyset$. We impose the induction hypothesis that $\mathbf{H}_{X_0}^{\text{sph}}[r] \cong \hat{\mathbf{u}}_X^+[r]$ holds for $r < \text{rk}(\alpha)$, the pieces of rank strictly smaller than α . We look at the case when $\text{supp}(\mathbf{P}) \cap \mathbf{Coh}_\alpha^{\text{as}} \neq \emptyset$. Using Lemma 4.12 and similar argument as above, we can find $\underline{\beta}$ not very unstable such that $\text{supp}(\mathbf{P}) \subseteq \overline{\text{HN}^{-1}(\underline{\beta})}$. We assume $\underline{\beta} = (\beta_1, \dots, \beta_s)$, then by assumption there exists $1 \leq i \leq s-1$ such that $\mu(\beta_{i+1}) - \mu(\beta_i) > 2g_X - 2$. Apart from i , we separate $\underline{\beta}$ into two parts $\underline{\beta}^{\leq i}$ and $\underline{\beta}^{> i}$ with $\underline{\beta}^{\leq i} = \sum_{j=1}^i \beta_j$, $\underline{\beta}^{> i} = \sum_{j=i+1}^s \beta_j$. We write $\text{CT}_{\underline{\beta}^{\leq i}, \underline{\beta}^{> i}}^{\text{Coh}}(\mathbf{P})$ as $\mathbf{P}_1 \boxtimes \mathbf{P}_2$, where $\mathbf{P}_1$ and $\mathbf{P}_2$ are supported in the stacks $\mathbf{Coh}_{\underline{\beta}^{\leq i}}$ and $\mathbf{Coh}_{\underline{\beta}^{> i}}$ respectively. By induction hypothesis, $f_{\mathbf{P}_i} \in \mathbf{H}_{X_0}^{\text{sph}}$ for $i = 1, 2$. Note that

$\mu(\beta^{>i}) - \mu(\beta^{\leq i}) \geq \mu(\beta_{i+1}) - \mu(\beta_i)$, thus the HN-type given by $(\beta^{\leq i}, \beta^{>i})$ is very unstable. We can apply Lemma 4.13 to deduce that $\mathbf{P} \subseteq \mathbf{R} := \text{Eis}_{\beta^{\leq i}, \beta^{>i}}^{\text{Coh}}(\mathbf{P}_1 \boxtimes \mathbf{P}_2)$ and the HN-types of simple components in the complementary of $\mathbf{P}$ in $\mathbf{R}$ are $< \underline{\beta}$. For another simple component $\mathbf{P}' \subseteq \mathbf{R}$, if its HN-type is very unstable, then we already know that $f_{\mathbf{P}'} \in \mathbf{H}_{X_0}^{\text{sph}, \text{vus}}$. Otherwise, we apply the previous steps for $\mathbf{P}$ to $\mathbf{P}'$, after several repeats, all the simple components created locates in very unstable locus and we obtain $f_{\mathbf{R}} - f_{\mathbf{P}} \in \mathbf{H}_{X_0}^{\text{sph}}$. Since we have also $\text{Eis}^{\text{Coh}}(f_{\mathbf{P}_1}, f_{\mathbf{P}_2}) \in \mathbf{H}_{X_0}^{\text{sph}}$ by induction hypothesis, we can deduce that $f_{\mathbf{P}} \in \mathbf{H}_{X_0}^{\text{sph}}$.

Finally, we deal with the case where $\text{supp}(\mathbf{P}) \cap \text{Coh}_{\alpha}^{\text{as}} \neq \emptyset$. We denote by $\mathbf{H}_{X_0}^{\text{sph}, \text{us}}[\alpha] \mathbf{H}_{X_0}^{\text{sph}}[\alpha]$ the subspace spanned by $f_{\mathbf{P}}$ such that $\text{supp}(\mathbf{P}) \cap \text{Coh}_{\alpha}^{\text{as}} = \emptyset$, which includes the functions considered in the above two cases. Now we want to show

$$\hat{\mathbf{u}}_X^+[\alpha] / \mathbf{H}_{X_0}^{\text{sph}, \text{us}}[\alpha] \cong \mathbf{H}_{X_0}^{\text{sph}}[\alpha] / \mathbf{H}_{X_0}^{\text{sph}, \text{us}}[\alpha]$$

We know $\hat{\mathbf{u}}_X^+[\alpha] / \mathbf{H}_{X_0}^{\text{sph}, \text{us}}[\alpha]$ is finite dimensional and admits a non-degenerate inner product inherit from $\hat{\mathbf{u}}_X^+[\alpha]$, it thus suffices to show for any $\bar{f} \in \hat{\mathbf{u}}_X^+[\alpha] / \mathbf{H}_{X_0}^{\text{sph}, \text{vus}}[\alpha]$, there exists $\bar{g} \in \mathbf{H}_{X_0}^{\text{sph}}[\alpha] / \mathbf{H}_{X_0}^{\text{sph}, \text{vus}}[\alpha]$ such that $\langle \bar{f}, \bar{g} \rangle \neq 0$.

In fact, it suffices to show for any $\mathbf{P} \in \mathcal{P}_{\alpha}$ such that $\text{supp}(\mathbf{P}) \cap \text{HN}^{-1}(\underline{\beta}) \neq \emptyset$ for some almost semi-stable HN-type $\underline{\beta}$, there exists $g \in \mathbf{H}_{X_0}^{\text{sph}}[\alpha]$ such that $\langle f_{\mathbf{P}}, g \rangle \neq 0$. By assumption, there exists $\mathbf{P}'$ such that $\mathbf{P} \oplus \mathbf{P}' = \text{Eis}_{\underline{\alpha}}^{\text{Coh}}(\mathbf{1}_{\underline{\alpha}})$ for some principal $\underline{\alpha}$, which is not necessarily HN-type. Since $\text{supp}(\mathbf{P}) \cap \text{HN}^{-1}(\underline{\beta}) \neq \emptyset$, the constant term $\text{CT}_{\underline{\beta}}^{\text{Coh}}(\mathbf{P}) \neq 0$. On the other hand, we have Langlands formula that

$$\text{CT}_{\underline{\beta}}^{\text{Coh}} \circ \text{Eis}_{\underline{\alpha}}^{\text{Coh}}(\mathbf{1}_{\underline{\alpha}}) = \bigoplus_{\underline{\gamma}} \text{Eis}_{\underline{\gamma}}^{\underline{\beta}, \text{Coh}} \circ \text{CT}_{\underline{\alpha}}^{\underline{\gamma}, \text{Coh}}(\mathbf{1}_{\underline{\alpha}})$$

where $\underline{\gamma}$ runs over the contingency matrices of $\underline{\alpha}$ and $\underline{\beta}$. Thus $\text{CT}_{\underline{\beta}}^{\text{Coh}}(\mathbf{P})$ is a component in $\bigoplus_{\underline{\gamma}} \text{Eis}_{\underline{\gamma}}^{\underline{\beta}, \text{Coh}} \circ \text{CT}_{\underline{\alpha}}^{\underline{\gamma}, \text{Coh}}(\mathbf{1}_{\underline{\alpha}})$. For any $g' \in \mathbf{H}_{X_0}^{\text{sph}}[\underline{\beta}]$, according to Lemma 4.3 (ii), we have

$$\langle f_{\mathbf{P}}, \text{Eis}_{\underline{\beta}}^{\text{Coh}}(g') \rangle = \langle \text{CT}_{\underline{\beta}}^{\text{Coh}}(f_{\mathbf{P}}), g' \rangle = \left\langle \sum_{\underline{\gamma}} \text{Eis}_{\underline{\gamma}}^{\underline{\beta}, \text{Coh}} \circ \text{CT}_{\underline{\alpha}}^{\underline{\gamma}, \text{Coh}}(\mathbf{1}_{\underline{\alpha}}), g' \right\rangle$$

Since $\underline{\alpha}$ is principal, its subpartition $\underline{\gamma}$ is again principal. We know $\text{CT}_{\underline{\alpha}}^{\underline{\gamma}, \text{Coh}} : \mathbf{H}_{X_0}^{\text{sph}}[\underline{\alpha}] \rightarrow \mathbf{H}_{X_0}^{\text{sph}}[\underline{\gamma}]$ preserves spherical part, thus the left side $\sum_{\underline{\gamma}} \text{Eis}_{\underline{\gamma}}^{\underline{\beta}, \text{Coh}} \circ \text{CT}_{\underline{\alpha}}^{\underline{\gamma}, \text{Coh}}(\mathbf{1}_{\underline{\alpha}})$ in the pairing appears in $\mathbf{H}_{X_0}^{\text{sph}}[\underline{\beta}]$, we can take g' to be exactly the same function, this will give a non-vanishing pairing $\langle f_{\mathbf{P}}, \text{Eis}_{\underline{\beta}}^{\text{Coh}}(g') \rangle$ and clearly $\text{Eis}_{\underline{\beta}}^{\text{Coh}}(g')$ is the function in $\mathbf{H}_{X_0}^{\text{sph}}[\alpha]$ we want. This finishes the proof.

5 Singular support of Eisenstein sheaves

5.1 Irreducible nilpotent Higgs sheaves

We denote by $\mathbf{Higgs}_\alpha$ the moduli stack of Higgs sheaves on $\mathbf{Coh}_\alpha$, i.e. sheaf in groupoid given by

$$S \mapsto \left\langle (\mathcal{F}, \varphi) \mid \mathcal{F} \in \mathbf{Coh}_\alpha(S), \varphi \in H^0 \left(X_S, \text{End}(X_S) \otimes \Omega_{X_S}^1 \right) \right\rangle$$

where φ is called the Higgs field. One can see that $\mathbf{Higgs}_\alpha$ is the cotangent space $T^*\mathbf{Coh}_\alpha$ to $\mathbf{Coh}_\alpha$. Restrict to those φ nilpotent, we get $\Lambda_\alpha \subseteq \mathbf{Higgs}_\alpha$, called the global nilpotent cone, come from [Lau88]. Moreover, it is well-known that Λ_α is a Lagrangian in $\mathbf{Higgs}_\alpha$. Unlike $\mathbf{Coh}_\alpha$, which is irreducible, the Lagrangian Λ_α has many irreducible components. Those irreducible components of Λ_α can be arranged combinatorially via Jordan stratification, which is described explicitly in [Boz22], we give here a quick review.

We first investigate $(\mathcal{F}, \varphi) \in \Lambda_\alpha$, since φ is nilpotent, there exists minimal $s \in \mathbb{N}$ such that $\varphi^s = 0$ and $\varphi^{s-1} \neq 0$. Let $\mathcal{F}_i := \text{Im}(\varphi^i) \otimes \Omega_X^{-i}$ and put $\mathcal{F}_0 := \mathcal{F}$, this gives us two series:

$$\begin{aligned} \mathcal{F}_0 &\rightarrow \mathcal{F}_1 \otimes \Omega_X^1 \rightarrow \mathcal{F}_2 \otimes \Omega_X^2 \rightarrow \dots \rightarrow \mathcal{F}_s \otimes \Omega_X^s = 0 \\ \mathcal{F}_0 &\hookleftarrow \mathcal{F}_1 \hookleftarrow \mathcal{F}_2 \hookleftarrow \dots \hookleftarrow \mathcal{F}_s = 0 \end{aligned}$$

The second series allows us to define $\mathcal{F}_k'' := \mathcal{F}_k / \mathcal{F}_{k+1}$ and the first series induces

$$\mathcal{F}_0'' \rightarrow \mathcal{F}_1'' \otimes \Omega_X^1 \rightarrow \mathcal{F}_2'' \otimes \Omega_X^2 \rightarrow \dots \rightarrow \mathcal{F}_s'' \otimes \Omega_X^s = 0$$

We abbreviate $\mathcal{F}_k'' \otimes \Omega_X^k$ as $\mathcal{F}_k''(k\Omega_X)$ and denote by α_k the class of $\text{Ker} \left(\mathcal{F}_{k-1}''((k-1)\Omega_X) \rightarrow \mathcal{F}_k''(k\Omega_X) \right)$, we use $\alpha_k(m\Omega)$ to denote the class of $\text{Ker} \left(\mathcal{F}_{k-1}''((k-1)\Omega) \rightarrow \mathcal{F}_k''(k\Omega_X) \right)(m\Omega_X)$, the series $\alpha = (\alpha_1, \dots, \alpha_s)$ is called the *Jordan type* of $(\mathcal{F}, \varphi)$, corresponds to the outer blocks of the following Young tableau:

α_s				
$\alpha_s(-\Omega)$	α_{s-1}			
...	...	...	$\ddots$	
$\alpha_s((1-s)\Omega)$	$\alpha_{s-1}((2-s)\Omega)$	...	$\alpha_2(-\Omega)$	α_1

All blocks in the Young tableau sums to α , thus tuple $\alpha = (\alpha_1, \dots, \alpha_s)$ corresponds to $(\mathcal{F}, \varphi)$ should satisfy the following relations:

$$\sum_{k=1}^s k \cdot \text{rk}(\alpha_k) = \text{rk}(\alpha); \quad \sum_{k=1}^s k \cdot \deg(\alpha_k) - \text{rk}(\alpha_k) \left(\sum_{j=1}^{k-1} j \cdot (2g-2) \right) = \deg(\alpha)$$

For any tuple with coordinates in $\mathbf{Z}^+$ which is potentially a Jordan type of nilpotent Higgs sheaf of class α should satisfy the above condition. If α is such, we say α is a Jordan type of α and use the notation $\alpha \vdash \alpha$ in [Boz22]. We have a map $\text{JT} : \Lambda_\alpha \longrightarrow \{\text{Jordan types of } \alpha\}$.

Theorem 5.1. [Boz22, Corollary 2.5] For $\alpha \vdash \alpha$, we denote by $\Lambda_\alpha \subseteq \Lambda_\alpha$ the preimage $\text{JT}^{-1}(\alpha)$. Then this gives a cell decomposition $\Lambda_\alpha = \bigsqcup_{\alpha \vdash \alpha} \Lambda_\alpha$. Moreover, the set of irreducible components of Λ_α is given by

$$\left\{ \overline{\Lambda_\alpha} \mid \alpha \vdash \alpha \right\}, \quad \text{where } \overline{\Lambda_\alpha} \text{ is the Zariski closure of } \Lambda_\alpha$$

For Jordan type $\alpha = (\alpha_1, \dots, \alpha_s)$, there is a projection $\pi_\alpha : \Lambda_\alpha \longrightarrow \prod_{k=1}^s \mathbf{Coh}_{\alpha_s}$ given by

$$\pi_\alpha : (\mathcal{F}, \varphi) \mapsto (\text{Ker}(\mathcal{F}_{k-1}''((k-1)\Omega_X) \rightarrow \mathcal{F}_k''(k\Omega_X)))$$

which is an iterated vector bundle, hence Λ_α is irreducible, to show that $\overline{\Lambda_\alpha}$ is exactly the irreducible component of Λ_α , it is enough to calculate its dimension, which is done in [Boz22]. Moreover, we can describe the closure $\overline{\Lambda_\alpha}$ via a partial order on Jordan type of α . For two Jordan types $\alpha = (a_1, \dots, a_s)$ and $\beta = (\beta_1, \dots, \beta_t)$ of α , we say $\beta \leq \alpha$ if the following holds:

$$\sum_{h=1}^k \sum_{j=h}^s \alpha_j ((h-j)(2g-2)) \leq \sum_{h=1}^k \sum_{j=h}^t \beta_j ((h-j)(2g-2))$$

where for each k , the sum is over blocks in the first k rows (from bottom to top) in the Young tableau and the relation $\leq$ is the one in $\mathbf{Z}^+$. Thus the minimal Jordan type of α is itself under this order, we denote this Jordan type by (α) , in which case $(\mathcal{F}, \varphi) \in \Lambda_{(\alpha)}$ is the zero section $\mathbf{Coh}_\alpha \subseteq \mathbf{Higgs}_\alpha$. In general, the closed irreducible components of Λ_α can be described as $\overline{\Lambda_\alpha} = \bigsqcup_{\beta \leq \alpha} \Lambda_\beta$.

Similar to the diagram of Eis^{Coh} and CT^{Coh} , we have induction diagram on $\mathbf{Higgs}$:

$$\begin{array}{ccccc} \mathbf{Higgs}_\alpha \times \mathbf{Higgs}_\beta & \xleftarrow{q_{\alpha,\beta}^H} & \widetilde{\mathbf{Higgs}}_{\alpha,\beta} & \xrightarrow{p_{\alpha,\beta}^H} & \mathbf{Higgs}_{\alpha+\beta} \\ \uparrow & & \uparrow & & \uparrow \\ \Lambda_\alpha \times \Lambda_\beta & \xleftarrow{q_{\alpha,\beta}^\Lambda} & \widetilde{\Lambda}_{\alpha,\beta} & \xrightarrow{p_{\alpha,\beta}^\Lambda} & \Lambda_{\alpha+\beta} \end{array}$$

where $\widetilde{\mathbf{Higgs}}_{\alpha,\beta}$ is the derived stack classifying following commutative diagrams:

$$\left\langle \begin{array}{ccccc} \mathcal{E}' & \xrightarrow{f} & \mathcal{E}'' & \xrightarrow{g} & \mathcal{E} \\ \downarrow \phi' & & \downarrow \phi'' & & \downarrow \phi \\ \mathcal{E}' \otimes \Omega_X^1 & \xrightarrow{f \otimes \Omega_X^1} & \mathcal{E}'' \otimes \Omega_X^1 & \xrightarrow{g \otimes \Omega_X^1} & \mathcal{E} \otimes \Omega_X^1 \end{array} \right| \begin{array}{l} (\mathcal{E}, \phi) \in \mathbf{Higgs}_\alpha; (\mathcal{E}', \phi') \in \mathbf{Higgs}_\beta; \\ (f, g) \in \text{Ext}^1(\mathcal{E}, \mathcal{E}') \end{array} \right\rangle$$

and the maps $q_{\alpha,\beta}^H$ (resp. $p_{\alpha,\beta}^H$) are given by projection to the third and the first column (resp. the middle column). The stack $\widetilde{\Lambda}_{\alpha,\beta}$ in the bottom is the substack of $\widetilde{\mathbf{Higgs}}_{\alpha,\beta}$ by adding the condition that one (hence all) of ϕ', ϕ'', ϕ is nilpotent.

Proposition 5.2. *The maps $q_{\alpha,\beta}^H$ and $q_{\alpha,\beta}^\Lambda$ are quasi-smooth and the maps $p_{\alpha,\beta}^H$ and $p_{\alpha,\beta}^\Lambda$ are representable.*

As a corollary, we have the pull-push functors $\text{Eis}_{\alpha,\beta}^H := (p_{\alpha,\beta}^H)_! (q_{\alpha,\beta}^H)^*$ and $\text{Eis}_{\alpha,\beta}^\Lambda = (p_{\alpha,\beta}^\Lambda)_! (q_{\alpha,\beta}^\Lambda)^*$ induce morphisms:

$$\begin{aligned} \text{Eis}_{\alpha,\beta}^H &: H_\bullet^{\text{BM}}(\mathbf{Higgs}_\alpha \times \mathbf{Higgs}_\beta) \longrightarrow H_\bullet^{\text{BM}}(\mathbf{Higgs}_{\alpha+\beta}) \text{ and} \\ \text{Eis}_{\alpha,\beta}^\Lambda &: H_\bullet^{\text{BM}}(\Lambda_\alpha \times \Lambda_\beta) \longrightarrow H_\bullet^{\text{BM}}(\Lambda_{\alpha+\beta}) \end{aligned}$$

In particular $\text{Eis}_{\alpha,\beta}^\Lambda([\Lambda_{(\alpha)}] \boxtimes [\Lambda_{(\beta)}])$ contains $[\Lambda_{(\alpha+\beta)}]$ (with multiplicity 1).

We can see the induction structure induces a multiplication structure on $\bigoplus_\alpha H_\bullet^{\text{BM}}(\mathbf{Higgs}_\alpha)$, respectively on $\bigoplus_\alpha H_\bullet^{\text{BM}}(\Lambda_\alpha)$. The resulting algebras are referred as *cohomological Hall algebra* (CoHA in short) of Higgs sheaves (nilpotent Higgs sheaves respectively). Moreover, there is a co-multiplication structure given by (co)factorization structure.

Theorem 5.3 (Kinjo et al.). *There is an isomorphism $H_\bullet^{\text{BM}}(\mathbf{Higgs}_\alpha) \cong H_\bullet(\mathbf{Higgs}_\alpha^{\text{an}})$, where the right-hand side is the (singular) homology of the analytic stack $\mathbf{Higgs}_\alpha^{\text{an}}$.*

*Moreover, there are comultiplication structures on $\bigoplus_\alpha H_\bullet^{\text{BM}}(\mathbf{Higgs}_\alpha)$, $\bigoplus_\alpha H_\bullet^{\text{BM}}(\Lambda_\alpha)$, $\bigoplus_\alpha H_{\text{top}}^{\text{BM}}(\Lambda_\alpha)$ given by the co-factorization of $T^*X \times \mathbb{A}^1$, and an analog of Langlands formula holds.*

5.2 Characteristic cycles on stacks

Next, we consider the characteristic cycle of Eisenstein sheaves.

Let first consider X a smooth scheme, with cotangent bundle T^*X . We say a Zariski closed subset $C \subseteq T^*X$ is a conical subset if it is invariant under $\mathbb{G}_m$ -action. Let $f : W \rightarrow X$ be a morphism of smooth schemes and $C \subseteq T^*X$ be a conical subset, we say f is C -transversal (resp. at a point w) if $(W \times_X C) \cap df^{-1}(T_w^*W)$ is included in the zero section $W \times_X T_X^*X \subseteq W \times_X T^*X$ (resp. the intersection with point w : $((W \times_X C) \cap df^{-1}(T_w^*W)) \times_W w$ is included in $W \times_X T_X^*X$).

In particular, when f is C -transversal, the image $df((W \times_X C) \cap df^{-1}(T_w^*W))$ is a conical subset in

T^*W . Likewise, for $h : X \rightarrow Y$ a morphism from smooth scheme X and $C \subseteq T^*X$ conical subset, we say h is C -transversal if $df^{-1}(C) \subseteq X \times_Y T_Y^*Y$. The pair of morphisms $X \xleftarrow{f} W \xrightarrow{h} Y$ with X, W smooth is said to be C -transversal if f is C -transversal and h is $df((W \times_X C) \cap df^{-1}(T_W^*W))$ -transversal.

Lemma 5.4. *Let $j : X_0 \hookrightarrow X_1$ be an open immersion between smooth irreducible schemes and let $\mathcal{F} \in \text{Shv}_{\text{constr}}(X_0)$ with characteristic cycle $\text{CC}(\mathcal{F}) = \sum_a m_a [\Lambda_a^0]$, then*

$$\text{CC}((j)_!(\mathcal{F})) = \sum_a m_a [\Lambda_a^1], \text{ where } \Lambda_a^1 \times_{X_1} X_0 = \Lambda_a^0$$

In other words, $(dj)^{-1}(\text{CC}(\mathcal{F})) \cong \text{CC}((j)_!(\mathcal{F}))$. We next consider the characteristic cycles for global quotient stack of the form $[X/G]$, where X is a smooth scheme over $\mathbf{k}$ and G is a reductive group over $\mathbf{k}$. Suppose we have diagram $Y \xleftarrow{q} V \xrightarrow{p} X$ where Y, V admits action of $P \subseteq G$ subgroup, X admits an action of G and that q is P -equivariant smooth map, p is closed P -equivariant. Then we have naturally the diagram $Y \times^P G \leftarrow V \times^P G \rightarrow X$. Quotienting G of the diagram, we get $[Y/P] \xleftarrow{\bar{q}} [V/P] \xrightarrow{\bar{p}} [X/G]$, compose $\bar{q}$ with $U(P)$ -gerbe $[Y/P] \rightarrow [Y/L]$, where L is the Levi subgroup of P and $U(P)$ is the unipotent radical of P , we get a correspondence diagram:

$$[Y/L] \xleftarrow{\tilde{q}} [V/P] \xrightarrow{\bar{p}} [X/G]$$

where $\tilde{q}$ is smooth and $\bar{p}$ is proper, we denote $\text{Ind}_{[Y/L]}^{[X/G]} := (\bar{p})_!(\tilde{q})^*[\text{shift}] : D([Y/L]) \rightarrow D([X/G])$, the equivariant induction, where the shift is given by the relative dimension $\dim[V/P] - \dim[Y/L]$. The $\mathcal{D}$ -modules or constructible sheaves on $[Y/L]$ are L -equivariant ones on Y . We define the equivariant characteristic cycle on $[Y/L]$ to be the same as the characteristic cycle of the sheaf over underlying Y . In other words, put $\text{Fgt} : D([Y/L]) \rightarrow D(Y)$, then $\text{CC}(\mathcal{F}) := \text{CC}(\text{Fgt}(\mathcal{F}))$ for $\mathcal{F} \in D([Y/L])$. Then it is proved in [Hen24, Theorem 6.3] the following compatibility for the induction diagram $(\tilde{q}, \bar{p})$ above.

Proposition 5.5. *The equivariant induction is compatible with equivariant characteristic cycle, namely*

$$\text{CC}\left(\text{Ind}_{[Y/L]}^{[X/G]}(\mathcal{F})\right) \cong \text{Ind}_{[Y/L]}^{[X/G]}(\text{CC}(\mathcal{F}))$$

Also, if $j : Y_0 \hookrightarrow Y_1$ is G -equivariant open embedding of smooth irreducible schemes which induces $\bar{j} : [Y_0/G] \hookrightarrow [Y_1/G]$, we have $(dj)^{-1}(\text{CC}(\mathcal{F})) \cong \text{CC}((\bar{j})_!(\mathcal{F}))$.

In the spirit of Lemma 5.4 and the above generalization to global quotient. We assume that the stack $\mathcal{X}$ is irreducible, locally of finite type and can be written as colimit of quotient stacks $\mathcal{X} := \text{colim}_i [X_i/G_i]$, such that the colimit is taken along the arrows $[X_i/G_i] \hookrightarrow [X_j/G_j]$ whenever $[X_j/G_j] \cong [X_j^i/G_i]$ and $X_i \hookrightarrow X_j^i$. Then, for $\mathcal{F} := \text{colim}_i \mathcal{F}_i$ as a colimit of intermediate extension along open inclusions as

above, we can define the following *inductive characteristic cycle* of $\mathcal{F}$ on X :

$$\mathrm{CC}^{\mathrm{ind}}(\mathcal{F}) := \sum_a m_a [\Lambda_a], \quad \text{where } \Lambda_a \text{ runs over } \mathrm{Lagr}(T^*X)$$

In fact, under the above assumption, we can restrict to any of $\mathcal{F}_i$, then $\mathrm{CC}(\mathcal{F}_i)$ is of the form $\sum_a m_a^i [\Lambda_a^i]$, where $\Lambda_a^i = \Lambda_a \times_X [X_i/G_i]$ is the restriction of the irreducible Lagrangian Λ_a .

To work in the case of ℓ -adic sheaves, we adapt to Saito's construction of characteristic cycle.

Theorem 5.6 (Saito). *For $\mathcal{F} \in \mathrm{Shv}_{\mathrm{constr}}(X)$ over smooth irreducible scheme X and conical Lagrangian subset $C \subseteq T^*X$ there exists $\mathbb{Z}$ -linear combination $\mathrm{CC}_C(\mathcal{F}) = \sum_a m_a [C_a]$ over irreducible components of C such that for any C -transversal pair $X \xleftarrow{f} W \xrightarrow{h} Y$ that h is étale and Y a smooth curve, we have the following Milnor formula*

$$-\dim \Phi_u(f^*\mathcal{F}, h) = (\mathrm{CC}_C(\mathcal{F}), dh)_{T^*W, u}$$

for every at most C -isolated point $u \in W$. Moreover $\mathrm{CC}_C(\mathcal{F})$ is independent of conical subset $\mathcal{F}$ has micro-support on, we thus simplify as $\mathrm{CC}(\mathcal{F})$. The support $\mathrm{supp}(\mathrm{CC}(\mathcal{F})) = \bigcup_{m_a \neq 0} C_a$ is exactly $\mathrm{SS}(\mathcal{F})$.

Conj: we still have the above consequence for $\mathrm{CC}^{\mathrm{ind}}$

5.3 local chart

We consider the $\mathrm{Eis}^{\mathrm{Coh}}$ on local charts. Recall that $\mathbf{Coh}_\alpha$ can be exhibited as colimit $\mathrm{colim}_{\mathcal{L}} \mathbf{Coh}_\alpha^{>\mathcal{L}}$. For two classes α and β , we can consider $\mathbf{Coh}_\alpha^{>\mathcal{L}} \times \mathbf{Coh}_\beta^{>\mathcal{L}}$, which is isomorphic to $[\mathcal{Q}_\alpha^\mathcal{L}/G_\alpha^\mathcal{L}] \times [\mathcal{Q}_\beta^\mathcal{L}/G_\beta^\mathcal{L}]$.

Lemma 5.7. *For fixed $\mathcal{L}'$, there exists small enough $N \ll 0$ such that for any $\deg(\mathcal{L}) \leq N$, we have in $\widetilde{\mathbf{Coh}}_{\alpha, \beta}$.*

$$\left(\mathfrak{p}_{\alpha, \beta}^{\mathrm{Coh}}\right)^{-1} \left(\mathbf{Coh}_{\alpha+\beta}^{>\mathcal{L}'}\right) \subseteq \left(\mathfrak{q}_{\alpha, \beta}^{\mathrm{Coh}}\right)^{-1} \left(\mathbf{Coh}_\alpha^{>\mathcal{L}} \times \mathbf{Coh}_\beta^{>\mathcal{L}}\right)$$

This allows us to define locally the functor $\mathrm{Eis}^{\mathrm{Coh}}$. For $(\mathcal{L}, \mathcal{L}')$ fixed pair of line bundles that satisfy the above lemma, we starts with $\mathbf{Coh}_\alpha^{>\mathcal{L}} \times \mathbf{Coh}_\beta^{>\mathcal{L}}$ and we want to arrive at $\mathbf{Coh}_{\alpha+\beta}^{>\mathcal{L}'} \cong [\mathcal{Q}_{\alpha+\beta}^{\mathcal{L}, \mathcal{L}'} / G_{\alpha+\beta}^\mathcal{L}]$, where $G_\alpha^\mathcal{L} \times G_\beta^\mathcal{L}$ is a Levi in $G_{\alpha+\beta}^\mathcal{L}$. To make a parabolic induction diagram, we can consider $\widetilde{\mathcal{Q}}_{\alpha, \beta}^{\mathcal{L}, \mathcal{L}'} := \widetilde{\mathcal{Q}}_{\alpha, \beta}^\mathcal{L} \cap \mathcal{Q}_{\alpha+\beta}^{\mathcal{L}, \mathcal{L}'}$, where $\widetilde{\mathcal{Q}}_{\alpha, \beta}^\mathcal{L}$ is a subscheme of $\mathcal{Q}_{\alpha+\beta}^\mathcal{L}$ that classifies

$$\left\{ \left[\phi : \mathcal{L} \otimes \mathbf{k}^{\langle c(\mathcal{L}), \alpha+\beta \rangle} \twoheadrightarrow \mathcal{F} \right] \in \mathrm{Quot}_\alpha^\mathcal{L} \mid \mathrm{Im} \left(\phi|_{\mathcal{L} \otimes \mathbf{k}^{\langle c(\mathcal{L}), \beta \rangle}} \right) \text{ strongly generated by } \mathcal{L} \right\}$$

then $\widetilde{\mathcal{Q}}_{\alpha, \beta}^{\mathcal{L}, \mathcal{L}'}$ is closed in $\mathcal{Q}_{\alpha+\beta}^{\mathcal{L}, \mathcal{L}'}$ and carries an action of the parabolic $P_{\alpha, \beta}^\mathcal{L} = (G_\alpha^\mathcal{L} \times G_\beta^\mathcal{L}) \times U$. We denote by $\widetilde{\mathbf{Coh}}_{\alpha, \beta}^{\mathcal{L}, \mathcal{L}'} := [\widetilde{\mathcal{Q}}_{\alpha, \beta}^{\mathcal{L}, \mathcal{L}'} / P_{\alpha, \beta}^\mathcal{L}]$ the global quotient, then it is the stack classifying extensions

$$0 \rightarrow \mathcal{E}' \rightarrow \mathcal{E}'' \rightarrow \mathcal{E} \rightarrow 0; \quad \text{where } (\mathcal{E}, \mathcal{E}') \in \mathbf{Coh}_\alpha^{>\mathcal{L}} \times \mathbf{Coh}_\beta^{>\mathcal{L}}, \text{ and } \mathcal{E}'' \in \mathbf{Coh}_{\alpha+\beta}^{>\mathcal{L}'}$$

In conclusion, we have the following commutative diagram:

$$\begin{array}{ccccc}
\mathbf{Coh}_\alpha^{>\mathcal{L}} \times \mathbf{Coh}_\beta^{>\mathcal{L}} & \xleftarrow{q_{\alpha,\beta}^{\mathbf{Coh},\mathcal{L},\mathcal{L}'}} & \widetilde{\mathbf{Coh}}_{\alpha,\beta}^{\mathcal{L},\mathcal{L}'} & \xrightarrow{p_{\alpha,\beta}^{\mathbf{Coh},\mathcal{L},\mathcal{L}'}} & \mathbf{Coh}_{\alpha+\beta}^{>\mathcal{L}'} \\
\downarrow & & \downarrow & & \downarrow \\
\mathbf{Coh}_\alpha \times \mathbf{Coh}_\beta & \xleftarrow{q_{\alpha,\beta}^{\mathbf{Coh}}} & \widetilde{\mathbf{Coh}}_{\alpha,\beta} & \xrightarrow{p_{\alpha,\beta}^{\mathbf{Coh}}} & \mathbf{Coh}_{\alpha+\beta}
\end{array}$$

where the vertical arrows are open immersions. We denote by $\text{Eis}_{\alpha,\beta}^{\mathbf{Coh},\mathcal{L},\mathcal{L}'}$ for $\left(p_{\alpha,\beta}^{\mathbf{Coh},\mathcal{L},\mathcal{L}'}\right)_! \left(q_{\alpha,\beta}^{\mathbf{Coh},\mathcal{L},\mathcal{L}'}\right)^*$, under the previous notation, this is exactly $\text{Ind} \left[\frac{Q_{\alpha+\beta}^{\mathcal{L},\mathcal{L}'} / G_{\alpha+\beta}^{\mathcal{L}}}{Q_\alpha^{\mathcal{L}} \times Q_\beta^{\mathcal{L}} / G_\alpha^{\mathcal{L}} \times G_\beta^{\mathcal{L}}} \right]$. Then by Proposition 5.5, we have:

$$\text{CC} \left(\text{Eis}_{\alpha,\beta}^{\mathbf{Coh},\mathcal{L},\mathcal{L}'}(\mathcal{F}) \right) \cong \text{Eis}_{\alpha,\beta}^{\mathbf{Coh},\mathcal{L},\mathcal{L}'}(\text{CC}(\mathcal{F}))$$

Proposition 5.8. *The objects in $Q_\alpha^{\mathbf{Coh}}$ has nilpotent singular support.*

We consider $Q_\alpha^{\mathbf{Coh}}$ the (Karoubi complete) Laumon-Lusztig category of $\mathbf{Coh}_\alpha$. According to the local chart of CC, we can conclude

Proposition 5.9. *The inductive characteristic cycle CC^{ind} induces a map $K_0(Q_\alpha^{\mathbf{Coh}}) \longrightarrow H_{\text{top}}^{\text{BM}}(\Lambda_\alpha)$, which commutes with $\text{Eis}^{\mathbf{Coh}}$, namely we have the following commutative diagram:*

$$\begin{array}{ccc}
K_0(Q_\alpha^{\mathbf{Coh}}) \hat{\otimes} K_0(Q_\beta^{\mathbf{Coh}}) & \xrightarrow{\text{CC}^{\text{ind}}} & H_{\text{top}}^{\text{BM}}(\Lambda_\alpha \times \Lambda_\beta) \\
\text{Eis}_{\alpha,\beta}^{\mathbf{Coh}} \downarrow & & \downarrow \text{Eis}_{\alpha,\beta}^\Lambda \\
K_0(Q_{\alpha+\beta}^{\mathbf{Coh}}) & \xrightarrow{\text{CC}^{\text{ind}}} & H_{\text{top}}^{\text{BM}}(\Lambda_{\alpha+\beta})
\end{array}$$

In other words, CC^{ind} is an algebra morphism between $K_0(Q^{\mathbf{Coh}})$ and $H_{\text{top}}^{\text{BM}}(\Lambda)$.

Theorem 5.10. *The inductive characteristic cycle CC^{ind} is compatible with the co-algebra structures on two sides, namely we have the following commutative diagram:*

$$\begin{array}{ccc}
K_0(Q_\alpha^{\mathbf{Coh}}) & \xrightarrow{\text{CC}^{\text{ind}}} & H_{\text{top}}^{\text{BM}}(\Lambda_\alpha) \\
\text{CT}_{\beta,\gamma}^{\mathbf{Coh}} \downarrow & & \downarrow \hat{\Delta}_{\beta,\gamma} \\
K_0(Q_\beta^{\mathbf{Coh}}) \hat{\otimes} K_0(Q_\gamma^{\mathbf{Coh}}) & \xrightarrow{\text{CC}^{\text{ind}}} & H_{\text{top}}^{\text{BM}}(\Lambda_\beta \times \Lambda_\gamma)
\end{array}$$

Moreover, CC^{ind} establishes an isomorphism of bialgebras $K_0(Q^{\mathbf{Coh}}) \cong H_{\text{top}}^{\text{BM}}(\Lambda)$.

Proof. According to Theorem 4.4, it is enough to prove for $\text{Eis}_{\underline{\alpha}}^{\mathbf{Coh}}(\mathbf{1}_{\underline{\alpha}})$ for all partitions $\underline{\alpha}$ of α which are

principal. Thus we need to calculate the upper circuit of the diagram:

$$\hat{\Delta}_{\beta,\gamma} \left(\text{CC}^{\text{ind}} \left(\text{Eis}_{\underline{\alpha}}^{\text{Coh}} (\mathbf{1}_{\underline{\alpha}}) \right) \right) = \hat{\Delta}_{\beta,\gamma} \left(\text{Eis}_{\underline{\alpha}}^{\Lambda} (\text{CC}^{\text{ind}} (\mathbf{1}_{\underline{\alpha}})) \right)$$

where the equivalence is given by Proposition 5.9. We notice that $\text{CC}^{\text{ind}} (\mathbf{1}_{\underline{\alpha}}) = [\Lambda_{(\alpha_1)}] \otimes \dots \otimes [\Lambda_{(\alpha_s)}]$. Thus the top circuit of the diagram (applying to Eis^{Coh}) is

$$\hat{\Delta}_{\beta,\gamma} \left(\text{Eis}_{\underline{\alpha}}^{\Lambda} ([\Lambda_{(\alpha_1)}] \otimes \dots \otimes [\Lambda_{(\alpha_s)}]) \right)$$

On the other hand, the lower circuit gives

$$\begin{aligned} \text{CC}^{\text{ind}} \left(\text{CT}_{\beta,\gamma}^{\text{Coh}} \circ \text{Eis}_{\underline{\alpha}}^{\text{Coh}} (\mathbf{1}_{\underline{\alpha}}) \right) &= \text{CC}^{\text{ind}} \left(\bigoplus_{\underline{\beta}, \underline{\gamma}} \text{Eis}_{\underline{\beta}}^{\text{Coh}} (\mathbf{1}_{\underline{\beta}}) \boxtimes \text{Eis}_{\underline{\gamma}}^{\text{Coh}} (\mathbf{1}_{\underline{\gamma}}) \right) \\ &= \bigoplus_{\underline{\beta}, \underline{\gamma}} \left(\text{Ind}_{\underline{\beta}} ([\Lambda_{(\beta_1)}] \otimes \dots \otimes [\Lambda_{(\beta_k)}]) \right) \otimes \left(\text{Ind}_{\underline{\gamma}} ([\Lambda_{(\gamma_1)}] \otimes \dots \otimes [\Lambda_{(\gamma_l)}]) \right) \end{aligned}$$

Thus the compatibility of CC^{ind} and CT^{Coh} is a consequence of the Langlands formula for $H_{\text{top}}^{\text{BM}}(\Lambda)$ (Theorem 5.3). ■

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